■ প্রশ্ন কমনের প্রমাণ পত্র ■

প্রমাণপত্র : ২০২৪ সালের পুরাতন সাজেশন্স থেকে কমনকৃত প্রশ্নের হার ৯৮%।

National University

[BBA (Hon's) 3rd Year; Exam.-2024 (Held in-04/01/2026)]

(Department of Marketing)

Subject Code : 232305

Subject : Financial Management (In English)

Time : 4 hours; **Full marks :** 80

[N.B. Answer different questions of each group chronologically.]

Group -A

1. Answer any **ten** questions : Marks 1 × 10 = 10

(a) What is financial management? *[অধ্যায়-০১ এর ০২ নং প্রশ্ন কমন পড়েছে]*

(b) What is rule of '69'? *[অধ্যায়-০২ এর ১৮ নং প্রশ্ন কমন পড়েছে]*

(c) What is trade credit? *[অধ্যায়-০৬ এর ১২ নং প্রশ্ন কমন পড়েছে]*

(d) What is right share? *[অধ্যায়-১০ এর ০৯ নং প্রশ্ন কমন পড়েছে]*

(e) What is CAPM? *[অধ্যায়-০৫ এর ১০ নং প্রশ্ন কমন পড়েছে]*

(f) What is capital rationing? *[অধ্যায়-০৩ এর ২০ নং প্রশ্ন কমন পড়েছে]*

(g) What is agency problem? *[অধ্যায়-০১ এর ১৪ নং প্রশ্ন কমন পড়েছে]*

(h) What is zero coupon bond? *[অধ্যায়-১০ এর ২০ নং প্রশ্ন কমন পড়েছে]*

(i) What is portfolio investment? *[অধ্যায়-০১ এর ০৭ নং প্রশ্ন কমন পড়েছে]*

(j) What is current ratio? *[অধ্যায়-০৪ এর ২০ নং প্রশ্ন কমন পড়েছে]*

(k) What is working capital? *[অধ্যায়-০৬ এর ০১ নং প্রশ্ন কমন পড়েছে]*

(l) What is capital market? *[অধ্যায়-১০ এর ১২ নং প্রশ্ন কমন পড়েছে]*

Group -B

[Answer any **five** questions.] Marks : 4 × 5 = 20

2. Make differences between wealth maximization and profit maximization. *[অধ্যায়-০১ এর ১০ নং প্রশ্ন কমন পড়েছে]*
3. *[অধ্যায়-০৬ এর Problem-12 নং (হুবহু) প্রশ্ন কমন পড়েছে]*
4. *[অধ্যায়-০৫ এর Problem-01 নং (হুবহু) প্রশ্ন কমন পড়েছে]*
5. *[অধ্যায়-০৪ এর Problem-06 নং (হুবহু) প্রশ্ন কমন পড়েছে]*
6. Distinguish between operating leverage and financial leverage. *[অধ্যায়-০৮ এর ০৭ নং প্রশ্ন কমন পড়েছে]*
7. *[অধ্যায়-০৬ এর Problem-16 নং (অনুরূপ) প্রশ্ন কমন পড়েছে]*
8. *[অধ্যায়-০৩ এর Problem-03 নং (অনুরূপ) প্রশ্ন কমন পড়েছে]*
9. *[অধ্যায়-০৮ এর Problem-15 নং (অনুরূপ) প্রশ্ন কমন পড়েছে]*

Group -C

[Answer any **five** questions.] Marks : 10 × 5 = 50

10. (a) Discuss brief the internal factors affecting financial decisions. *[অধ্যায়-০১ এর ০৫ নং প্রশ্ন কমন পড়েছে]*

(b) Discuss the scope of financial management. *[অধ্যায়-০১ এর ১১(খ) নং প্রশ্ন কমন পড়েছে]*

11. *[অধ্যায়-০৩ এর Problem-14 নং (হুবহু) প্রশ্ন কমন পড়েছে]*
12. *[অধ্যায়-০৪ এর Problem-06 নং (হুবহু) ইজিপ্সাস প্রশ্ন কমন পড়েছে]*
13. Discuss the factors to be considered in determining working capital. *[অধ্যায়-০৬ এর ০৬ নং প্রশ্ন কমন পড়েছে]*
14. *[অধ্যায়-০৮ এর Problem-03 নং (হুবহু) প্রশ্ন কমন পড়েছে]*
15. *[অধ্যায়-০২ এর Problem-20 নং (অনুরূপ) প্রশ্ন কমন পড়েছে]*
16. *[অধ্যায়-৭.২ এর Problem-04 নং (হুবহু) প্রশ্ন কমন পড়েছে]*
17. *[অধ্যায়-০৫ এর Problem-13 নং (হুবহু) ইজিপ্সাস প্রশ্ন কমন পড়েছে]*

National University

[BBA (Hon's) 3rd Year; Exam-2025 (Will be held-2026/27)]

Subject Code : 232305

Subject Name : Financial Management

[Baticrom latest Final Suggestions]

মার্কেটিং বিভাগের ছাত্রছাত্রীদের জন্য

Chapter 01 Nature of Financial Management

(আর্থিক ব্যবস্থাপনার প্রকৃতি)

Part-A (Brief Questions):

○○ 01. **What is finance (অর্থায়ন কি?)**

Ans : Finance can be defined as the art and science of managing money.

02. ***What is financial management?***
(আর্থিক ব্যবস্থাপনা কি?) [২০২৪ সালে যা এসেছিল]

Or, Define financial management
(আর্থিক ব্যবস্থাপনার সংজ্ঞা দাও।)

[N.U. 2012, 2014, 2017, 2020, 2022, 2024 (Marketing)]
[টেস্ট পরীক্ষা– কবি নজরুল, হাবিবুল্লাহ, রাজেন্দ্র ও যশোর সিটি কলেজ]

***Ans :** Financial management means planning, organizing, directing and controlling the financial activities such as procurement and utilization of funds of the enterprise.*

○○○03. **What is financing decision? 99%**
(অর্থসংস্থান সিদ্ধান্ত কি?)

Ans : Financing decision determines when, where and how to acquire funds to meet the firm's investment needs.

○○○04. **What is investment decision?**
(বিনিয়োগ সিদ্ধান্ত কি?) *[N.U. 2014]*

Ans : Investment decision involves the decision of allocating capital to long-term assets that are expected to generate benefits in the future.

○○ 05. **What is dividend decision? (লভ্যাংশ সিদ্ধান্ত কি?)**

Ans : Financial manager must decide whether the firm should distribute all its profits, retain them or distribution a portion and retain the balance. This decision is known as the dividend decision.

○○○06. **What is profit maximization?**
(মুনাফা সর্বাধিকরণ কি?)
[Test Exam : Dhaka, Titumir, Rajandroo, Adamjee cantonment College]

Ans : Profit maximization means maximizing the profit of the firm.

07. ***What is portfolio investment? [২০২৪ সালে যা এসেছিল]*** ***[N.U. (Marketing)2020, 2024]***

***Ans.** Portfolio investment refers to investments made in a group of assets, including transactions in equity securities, such as common stock, and debt securities, such as banknotes, bonds and debentures.*

○○○08 **What is EPS? 99%** *[N.U. (Marketing) 2021]*

Or, What are earnings per share (EPS)?
(শেয়ার প্রতি আয় কি?)

Ans : Earning per share is a financial metric that measures the portion of a company's profit that is allocated to each outstanding share of common stock.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○○09 **What is wealth maximization?**
(সম্পদ সর্বাধিকরণ কি?) 99%
[Test Exam :Titumir, Suhnawardy, Edward, & Rajendra College, N.U. 2012, 2015; (Marketing) 2016, 2017, 2019, 2021, 2023]

Or, Define wealth maximization.

Ans : Wealth maximization means maximizing the net present value or wealth of a course of action for the shareholders.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○ 10. **Who is treasurer? (ট্রেজার কি?)**

Ans : The treasurer is financial manager of a firm. The treasurer's functions are secondary and external works and relate to the liability side of the balance sheet.

○○○11. **Who is controller? (কন্ট্রোরাল কি?**

Ans : The controller is manager of a firm. The controller's functions are related with primary and internal works of the firms and concentrate the asset side of the balance sheet.

○○○(12) **What is meant by ethics?** 99%

[N.U. (Marketing) 2017]

Ans. The standards of conduct by which one's actions are judged as right of wrong, honest or dishonest, fair or unfair are ethics.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○○13. **What do you mean by business ethics? (ব্যবসায়ের নৈতিকতা বলতে কি বুঝায়?)**

[টেস্ট পরীক্ষা- ঢাকা, ইডেন, হাবিবুল্লাহ বাহার ও এসকে কলেজ]

Ans : The standards of conduct by which one's actions are judged as right of wrong, honest or dishonest, fair or unfair in the business environment.

14. What is agency problem? [২০২৪ সালে যা এসেছিল]
(এজেন্সি সমস্যা কি?)
[টেস্ট পরীক্ষা- ঢাকা কলেজ, ইডেন ও এসকে কলেজ; জা.বি. ২০১৫, ২০১৯, ২০২৪]
Ans : The probability of conflict of interest between the shareholders and management of a firm is known as agency problem.

○ 15. **What is financial market? (আর্থিক বাজার কি?)**

Ans : A financial market is market for creation and exchange of financial assets.

○○○16. **What is money market? (মুদ্রা বাজার কি?)**

[টেস্ট পরীক্ষা- কবি নজরুল কলেজ]

Ans : Money market is the market for securities with maturities of less than one year. Money market is the short-term market.

○○○17. **What is corporate governance?** 99%
(কর্পোরেট শাসন কি?)
[টেস্ট পরীক্ষা- ঢাকা, হাবীবুল্লাহ বাহার, যশোর সিটি ও এডওয়ার্ড কলেজ] [Test Exam :Dhaka & Eden College- N.U. 2014]
Or, Define corporate governance
[টেস্ট পরীক্ষা- হাবীবুল্লাহ, মহসিন ও রাজেন্দ্র কলেজ]

Ans : Corporate governance is the system, in which a corporation is governed and deals with the various interests of its consumers, shareholders, employees and society at large.

○○○(18) **What is business finance?** 99%
(ব্যবসায় অর্থায়ন কি?) *[N.U. (Marketing)-201*

Ans : Business finance may be defined as th process of raising, providing and managin of all the money to be used in connection wit business activities.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯ প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○ 19. **What is corporate finance?**
(কর্পোরেট অর্থায়ন কি?)
[Test Exam : Suhrawardy & Debendra Colleg

Ans : Corporate finance deals with promotio capitalization, financing, investing and financi administration of the corporation.

○○○20. **What is Hurdle rate?** *[টেস্ট পরীক্ষা- কবি নজরু সাতক্ষীরা ও যশোর সিটি কলেজ [N.U. 2015]*

Ans. Hudle rate is the minimum rate that a compan expects to earn when investing in a project.

○○○(21) **What is corporate social responsibility?**
Or, What is CSR? 99% *[টেস্ট পরীক্ষা- ইডে সাতক্ষীরা ও রাজেন্দ্র কলেজ; N.U. 2012, 2022]*

Ans : Corporate social responsibility is business outlook that acknowledges a firm responsibilities to its stakeholders and natura environment.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯ প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○ 22. **Who is a financial manager?**
(আর্থিক ব্যবস্থাপক কে?)

Ans : The person who plays the role to activa financial management is called financial manager.

○○○23. **What is risk-return trade-off?**
(ঝুঁকি-আয় সমন্বয় কী?)

Ans : Risk-return trade-off means to achiev profitability by reducing or avoiding risk.

○○ 24. **How many types of business finance ar there on the basis of duration?**
(মেয়াদের ভিত্তিতে ব্যবসায় অর্থায়ন কত প্রকার?)

Ans : There are three types of busines finance on the basis of duration-(i) Short-ter financing, (ii) Medium-term financing and (iii) Long-term financing.

○○○(25) **What is social responsibility?**
(সামাজিক দায়বদ্ধতা কী?) 99%
[N.U. Old 2013; (Marketing)-201

Ans : Social responsibility refers to the betterme of different interested group within a society.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯ প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○ 26. **What is a corporation?**
(অংশিদারী কারবার কাকে বলে?)

Ans : A corporation is an artificial bei created by law.

Part- B & C (Short & Broad Questions) :

○○ 01. What do you mean by finance? [Ans মার্কেটিং ইনসাইট (২য় খণ্ড) বুইয়ের 5 Page 1.01 No.]
Or, Defination of finance.

○○ 02. Define principles of financial management. [Ans : 6 Page 1.06 No.]
Or, Dicuss the principles of financial management.
[টেস্ট পরীক্ষা- রাজেন্দ্র, এডওয়ার্ড, যশোর সিটি কলেজ; N.U. 2017 (Finance-3rd)]

○○○ 03. Discuss the function of financial management? [Ans : 6 Page 1.07 No.]
[Test Exam : Dhaka, Eden, Mirpur, Kobi Nozzrul, Mohsin, Razedru, Rajandro College; N.U. BBS(Hon's) 2004, 2006; 2018 (Finance-3rd)]

○○○ 04. (a) Explain the concept of corporate governance. What are the principles of corporate governance?
[Ans : 11 Page 1.23+1.24 No.] *[টেস্ট পরীক্ষা- হাবীবুল্লাহ, সাদত ও রাজেন্দ্র কলেজ; N.U 2012; 2017 (Accounting-3rd)]*
(b) What should be maximized– profit or wealth? Explain? [Ans : 10 Page 1.20 No.]
[টেস্ট পরীক্ষা- ঢাকা, হাবিবুল্লাহ, রাজেন্দ্র, যশোর সিটি ও এডওয়ার্ড কলেজ] [N.U 2012]

○○○ (05) **Describe the scope of financial management.** 99% [Ans : Page No 7; Ques. No 1.08]
[N.U. (Marketing)-2016, 2018, 2022]
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○○ 06. **Briefly discuss the qualities of financial manager.** 99% [Ans : 8 Page 1.11 No.] *[N.U. Marketing-2016, 2018]*

○○○ (07) **What is agency problem? Discuss in brief the various techniques of overcoming agency problem.** 99%
[Ans : 9 + 10 Page 1.16 + 1.17 No.] *[টেস্ট পরীক্ষা- ইডেন, হাবিবুল্লাহ, বাঙলা, আযম খান কমার্স, যশোর সিটি ও এডওয়ার্ড কলেজ; N.U. 2009, 2012, 2013; 2017, 2019 (Marketing-3rd)]*
Or, Discuss in brief the various techniques of overcoming agency problem.
Or, What do you mean by agency problem? Discuss different market forces to minimize agency problem.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○○ 08. Does a financial manager have any social responsibility? If so, discuss. [Ans : Page No-9 ; Ques. No- 1.14]
Or, Discuss social responsibility of financial management. *[N.U. Marketing-2014, 2020]*

○○○ 09. **Why wealth maximization is considered as the ultimate goal of the firm? Explain.** 99%
[Ans : Page No-11 ; Ques. No- 1.21] *[N.U. 2014, 2018, 2020, 2021 (Marketing-3rd); 2018 (Managment-4th)]*
Or, "Wealth maximization, not the profit maximization is the ultimate objective of firm."—Explain.
Or, "Wealth maximization is considered as better measure than profit maximization" –Discuss.

○○○ 10. **(a) Discuss the importance of financial management.** 99% [Ans : Page No 7; Ques. No 1.09]
Or, Explain the significance of financial management in business. *[N.U. Marketing-2015, 2018, 2020, 2022]*
Or, Discuss the importance/Role/Necessity/advantages of financial management.

○○○ (11) **Discuss the goal and objectives of a financial manager.** [Ans : Page No 7; Ques. No 1.10]
[N.U. 2013, 2015; 2018 (Managment); 2017, 2019, 2021, 2023 (Marketing 3rd)]
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○○ (12) **(a) Discuss the principles of business finance.** 99% [Ans : Page No 5; Ques. No 1.04]
[N.U. 2017, 2019, 2023 (Marketing 3rd)]
(b) Discuss the three major decisions that financial manager must take. 99%
[Ans : Page No 8; Ques. No 1.13]
[টেস্ট পরীক্ষা- ঢাকা, এডওয়ার্ড, বিএম, তোলারাম কলেজ; N.U. Old 2013; (Managment-4th) 2017; 2018 (Accounting-3rd); 2016 (Finance-3rd) (Affiliated) 2016 (Accounting-3rd); 2015, 2017, 2019, 2021, 2023 (Marketing-3rd)]
Or, Discuss the three major decisions that financial manager must take.
Or, What are the major decisions to be taken by a financial manager? Explain.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

[২০২৪ সালের পরীক্ষায় যা এসেছিল–

2. *Make differences between wealth maximization and profit maximization.* **[Ans : Page No. 10; Ques. No 1.19]**
Or, Differentiate between wealth maximization and profit maximization. *[N.U. Marketing- 2016, 2022, 2024]*
Or, Distinguish between profit maximization and wealth maximization.

10. (a) *Discuss brief the internal factors affecting financial decisions.* **[Ans : Page No. 13; Ques. No 1.30]**
[N.U. 2012, 2014, 2016, 2024 (Marketing-3rd); 2017 (Accouting-3rd)]
Or, Discuss in brief the internal factors affecting financial decisions.
Or, Discuss the factors influencing financial decision,

(b) *Discuss the scope of financial management.* **[Ans : Page No. 7; Ques. No 1.08]**
[N.U. (Marketing)-2016, 2018, 2022, 2024]

Help Line *Dear Students, In above Suggestions Part- B and C : 05, 07, 11 & 12 No. Questions in are most important for the Examinee of 2025.* ..... *Editor*

Chapter 02 Time Value of Money
(অর্থের সময় মূল্য)

Part-A (Brief Questions) :

○○○01. **What is time value of money? (অর্থের সময় মূল্য কী?)** *[NU. BBA (Hons.)–2011]*
Ans : Time value of money refers to the concept that the same amount of money has a higher present value if received earlier rather than later.

○○ 02. **What are the concept of time value of money? (অর্থের সময় মূল্যের ধারণা কয়টি?)**
Ans : (i) Present value, (ii) Future Value.

○○○03 **What is annuity? (বার্ষিক বৃত্তি কী?)** 99%
[NU. (Marketing) 2023]
Ans : When equal amount of money is paid or, received continuously with equal time interval is called annuity.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○○04 **What is annuity due?** 99%
[NU. (Marketing) 2022]
Ans. An annuity whose payments occurs at the beginnings of each period is called an annuity due.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○ 05. **What are the factor of time value of money? (অর্থের সময় মূল্যের Factor গুলো লিখ।)**
Ans : (i) Time,
(ii) Interest rate,
(iii) Nature of interest rate,
(iv) Rate of infilation,
(v) Risk.

○○○06. **What do you mean by deferred annuity?** 99%
[D.U. (Affiliated) 2016]
Ans. If the annuity will be made often a specific period of time, it is called deferred annuity.

○○ 07. **What are the other names of future value? (ভবিষ্যৎ মূল্যের (Future value) অন্য নামগুলো কী কী?**
Ans : (i) Terminal value,
(ii) Compound Value.

○○○08. **What is compounding and discounting.**
[D.U. (Affiliated) (Marketing)2016]
Ans. Compounding : when we calculate interest on the interest and principal, it is called compounding.
Discounting : The ways of deducting interest from a compounding value is called discounting.

○○○09. **How many types of annuities are? (বার্ষিক বৃত্তি কত প্রকার)?**
Ans : Annuities are of four types
(a) ordinary annuity,
(b) Annuity due,
(c) Deferred annuity and
(d) Perpetual annuity.

○○○10. **What is meant by present value of money? (অর্থের বর্তমান মূল্য বলতে কী বুঝায়?)**
Ans : Present value is the current worth of a future amount of money that is discounted at an appropriate interest rate.

○○○11. **What is the elaboration of FVIF? (FVIF-এর পূর্ণরূপ কী?)** *[NU. 2012]*
Ans : FVIF = Future Value Interest Factor.

✪✪ 12. **What is the elaboration of FVIFA? (FVIFA-এর পূর্ণরূপ কী?)**

Ans : FVIFA = Future Value Interest Factor of an Annuity.

✪✪✪13. **What is the elaboration of PVIF? (PVIF-এর পূর্ণরূপ কী?)**

Ans : PVIF = Present Value Interest Factor.

✪✪✪14. **Write down the elaboration of PVIFA. (PVIFA-এর পূর্ণরূপ লিখ।)**

Ans : PVIFA = Present Value Interest Factor of an Annuity.

✪✪ 15. **Write down the formula of PVIFA. (PVIFA-এর সূত্রটি লিখ।)**

Ans : $\text{PVIFA} = \dfrac{1-\dfrac{1}{(1+i)^n}}{i}$

✪✪✪16. **What is the fundamental objective of time value of money? (অর্থের সময় মূল্যে ও মৌলিক উদ্দেশ্যে কোনটি?)**

Ans : The fundamental objective of time value of money is to make capital management modern and scientific.

✪✪✪ (17) **What is rule of 72? (৭২-এর বিধি কী?)** 99%

[NU (Marketing) 2017, 2019, 2023]

Ans : Rule of 72 gives a good approximation of the interest rate or, the number of years required to double the principal amount normally in case of annually compounding.

$i = \dfrac{72}{n}$ and $n = \dfrac{72}{i}$

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

18. What is rule of 69? (৬৯-এর বিধি কী?) [২০২৪ সালে যা এসেছিল]

[NU. BBA (Hon's)-2012, 2014; (Marketing) -2018, 2020, 2024]

***Ans :** Rule of 69 gives a good approximation of the interest rate or, the number of years required to double the principal amount normally in case of annual compounding. I* $= 0.35 + \dfrac{69}{n}$ $n = 0.35 + \dfrac{69}{i}$

✪✪ 19. **What is the another name of perpetual annuity? (চিরস্থায়ী বার্ষিক বৃত্তির অপর নাম কী?)**

Ans : Another name of perpetual annuity is perpetuity.

✪✪✪20. **What is time line?** 99%

[NU. BBA (Hon's)-(Marketing)-2018]

Ans. The line which shows the cash flows and its present or future value is called the time line.

✪✪✪ (21) **What is perpetuity?** 99%

[NU. BBA (Hon's) (Marketing)-2020]

Ans. A perpetuity is a security that providesa constant stream of identical cash flows indefinitely with no end.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪✪22. **What is compound interest?** 99%

[NU. BBA (Marketing)-2022]

Ans. Compound interest is the interest calculated on the principal and the interest accumulated over the previous period.

Part- B & C (Short & Broad Questions) :

✪✪✪ 01. (a) What is time value of money?
[Ans মার্কেটিং ইনসাইটস (২য় খণ্ড) বইয়ের 16 Page 2.01 No.]
(b) Discuss the limitation of time value of money. **[Ans : 17 Page 2.07 No.]**

✪✪✪ 02. **Explain the difference between present value and future value.** 99% **[Ans : 16 Page 2.03 No.]**
Or, Show the differences between present value and future value.

✪✪ 03. Discuss the Rule of 69. **[Ans : 17 Page 2.06 No.]**

✪✪ 04. What is sinking fund? **[Ans : 18 Page 2.10 No.]**

✪✪✪ (05) **Write down the reasons of time preference for money.** 99% **[Ans : 18 Page 2.08 No.]**

[N.U. (Marketing-3rd)-2019]

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪ 06. What is effective interest rate? **[Ans : 18 Page 2.11 No.]**

✪✪✪ 07. **What do you mean by annuity?** 99% **[Ans : 19 Page 2.15 No.]**
Or, Define annuity.

Help Line ***Dear Students, In above Suggestions Part- B and C : 05 No. Questions in are most Important for the Examinee of 2025.*** ***..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

Single Amount of PV & FV

ooo **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) (1st Year) Old 2010 No-9(a); 3rd Marketing 2021 No–03**

Mr. Ashraf/Mr. Alam deposited Tk. 5,00,000 in a bank. The bank promised to give 12% interest per annum. What will be the terminal value after 7 years if compounded annually?

Solution : প্র্যাক্টিক্যাল মার্কেটিং সাজেশন্স (২য় খণ্ড) বইয়ের ২১ পৃষ্ঠার **Problem-01** নং প্রশ্নের উত্তর দেখুন।

ooo (১১%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

ooo **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2019 No–06**

How much you have to deposit now to get Tk. 6,00,000 after 20 years at 8 percent interest per annum?

Solution : ২১ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Acc. (1st Year) 2014; 2nd Marketing 2014**

Nabiha Ibnat invests a single amount of Tk. 10,000 for 5 years at 10%. At the end of 5 years she takes the proceeds and invest them for 12 years at 15%. How much will she have after 17 years?

Solution : ২২ পৃষ্ঠার **Problem-03** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem–04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2020 No–03**

What rate is required for a sum of money to be triple after 5 years?

Solution : ২৫ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2016 No–04; 2023 No–06**

Mr. Ahnaf/Mr. Ashraf has taken a loan of Tk. 10,00,000 from a bank. After 10 years he has given the bank Tk. 17,00,000. Calculate the rate of interest if interest compounded quarterly.

Solution : ২৬ পৃষ্ঠার **Problem-11** নং প্রশ্নের উত্তর দেখুন।

ooo (১১%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

ooo **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2023 No–03**

In what time a sum of money will be doubled at 10% interest according to the rule 69?

Solution : ২৭ পৃষ্ঠার **Problem-13** নং প্রশ্নের উত্তর দেখুন।

ooo (১১%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

oo **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Acc. (1st Year) 2017**

Your father has offered you a choice one of the two following alternatives :—

(a) Tk 1,00,000 now.

(b) Tk 2,40,000 at the end of 8 year's.

Assume you could earn 10% annually. Which alternative should you choose and why?

Solution : ২৮ পৃষ্ঠার **Problem-16** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2017 No–11**

Find the future value of the following investments, time preference rate is 9% per year, compounded half yearly :

(i) Tk. 1,000 is invested each year beginning one year from now and continuing through year 10. When the proceeds are withdrawn

(ii) Tk. 1,000 is invested each year starting today and continuing through year 10. When the proceeds are withdrawn?

Solution : ২৯ পৃষ্ঠার **Problem-18** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Marketing 2018 No–11; 2022 No–03**

Calculate effective interest rate (EIR) for Tk. 5,000 invested in a project at 10% interest monthly.

Solution : ৩০ পৃষ্ঠার **Problem-21** নং প্রশ্নের উত্তর দেখুন।

০০০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

☑ PV & FV Annuity

○○ **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Finance (1st Year) 2016**

Your father committed to present you a car on your graduation after 7 years from now. The car will cost Tk. 35,00,000 at the time. Your father is planning to accumulate the fund by depositing equal amount of money in a bank account bearing 13% interest? How much should your father deposit in the bank account? Assume that deposit should be made in the beginning of the period. If the deposit is made at the end of the period then what will the amount of deposit?

Solution : ৩০ পৃষ্ঠার **Problem-22** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-11** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2023 No–15**

Your brother committed to present you a personal computer on your graduation after 5 years from now. The computer will cost Tk. 50,000 at that time. Your brother is planning to accumulate the fund by depositing equal amount of money in a bank account bearing 8% interest rate per annum. How much should your brother deposit in the bank account if the deposits are made (i) yearly, (ii) quarterly?

Solution : ৩১ পৃষ্ঠার **Problem-23** নং প্রশ্নের উত্তর দেখুন।

০০০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

○○ **Problem-12** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Acc. (1st Year) 2016**

Mr. Afnan borrows Tk. 60,000 at 12% interest and promises to pay off the loan in 20 annual installments beginning at the end of first year. What is the annual payment necessary?

Solution : ৩২ পৃষ্ঠার **Problem-24** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-13** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2023 No–04**

You have a plan to buy a motorcycle 5 years from now. The price of the motorcycle will be Tk. 5,00,000. You want to accumulate this money in a bank account by monthly installment, which gives you 10% interest. What is your monthly installment?

Solution : ৩৪ পৃষ্ঠার **Problem-29** নং প্রশ্নের উত্তর দেখুন।

০০০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

○○ **Problem-14** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Finance (1st Year) 2020 No-12(b)**

You wish to retire in 20 years, at which time you want to have accumulated enough money to receive an annual annuity of Tk. 19,000 for 26 years after retirement. During the period before retirement you can earn 12.25% annually, while after retirement you can earn 16% on your money.

What annual contributions to the retirement fund will allow you to receive the Tk. 19,000 annuity?

Solution : ৩৫ পৃষ্ঠার **Problem-32** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-15** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 2nd Mgt 2017 No-6**

You want to buy a motorcycle 4 years from now. The motorcycle will cost of Tk. 2,00,000. You have an opportunity to deposit NBL at 6% interest p.a. by monthly installment. What is your monthly installment?

Solution : ৩৭ পৃষ্ঠার **Problem-35** নং প্রশ্নের উত্তর দেখুন।

☑ PV & FV

ooo **Problem-16** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Marketing 2022 No–10**

You have a plan to accumulate Tk. 12,00,000 ten years from now for buying a car. You are deciding to deposit a certain amount of money in special account that will provide you 9% interest per annum.

(i)What will be your annual deposit at the end of each year?

(ii)What will be your annual deposit at the beginning of each year?

Solution : ৩৭ পৃষ্ঠার **Problem-37** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

ooo **Problem–17** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 1st Year Principles of Finance; Acc. 2014 No–13; BBA (Hons.) 3rd Marketing 2019 No–11**

You have a plan to accumulate Tk. 12,00,000 ten years from now for buying a car. You are deciding to deposit a certain amount of money in a special account that will provide you 9.5% interest per annum.

(i) What will be your annual deposit at the beginning of each year.

(ii) What will be your annual deposit at the end of year?

(iii) What will be your half-yearly deposit?

Solution : ৩৮ পৃষ্ঠার **Problem-38** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-18** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Acc. (1st Year) 2016, 2020 No-13**

Your father has promised to give you Tk. 1,00,000 in cash on your 25th birthday for a foreign tour. Today is your 16th birthday. He wants to know three things :

(a) If he decides to make annual payment into a fund after one year, how much will each have to be if the fund pays 8 percent?

(b) If he decides to invest a lump sum in the account after one year and let it compound annually, how much will the lump sum be?

(c) If in (a) the payments are made in the beginning of the year, how much it will be the value of annuity?

Solution : ৪০ পৃষ্ঠার **Problem-40** নং প্রশ্নের উত্তর দেখুন।

☑ Decision Making

oo **Problem-19** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Acc. (1st Year) 2015**

Your rich uncle has offered you a choice of one of the three following alternatives: Tk. 10,000 now; Tk. 2,000 a year for eight years or Tk. 24,000 at the end of eight years. Assuming you could earn 11% annually. Which alternative should you choose?

Solution : ৪১ পৃষ্ঠার **Problem-41** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem–20** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2019 No–12**

You are considering purchase of an apartment at Dhanmondi costing Tk. 1,00,00,000. You have three offers :

(i) Pay full in cash

(ii) Pay 25% in cash immediately and pay Tk. 13,20,000 each year for the next 10 years.

(iii) Pay 50% in cash and take a loan of Tk. 50,00,000 from BHBFC at 15% interest to be repaid in equal monthly instalment over the next 20 years. Assume 12% opportunity cost.

Which offer should you accept?

Solution : ৪৪ পৃষ্ঠার **Problem-43** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-]

⊗⊗ Problem-21 *Year Question & Solution* ⊙ N.U. BBA (Hon's) 3rd (Marketing) 2024 No–15

You are considering purchase of an apartment at Gulshan costing Tk. 1,00,00,000. You have three offers :

(i) Pay full in cash.

(ii) Pay 25% in cash immediately and pay Tk. 10,00,000 each year for the next 10 years.

(iii) Pay 50% in cash and take a loan of Tk. 50,00,000 from Sonali Bank PLC at 12% interest to be repaid in equal monthly installment over the next 15 years. Assume 10% opportunity cost.

Which offer should you accept?

Solution : ৫৪১ পৃষ্ঠার **Question No-15** নং প্রশ্নের উত্তর দেখুন।

ooo Problem–22 *Year Question & Solution* ⊙ N.U. BBA (Hons.) 3rd Marketing 2016 No–13; 2021 No–11

You are considering of an apartment in Dhaka costing Tk. 95,00,000.

You have three offers :

(i) Pay full in cash.

(ii) Pay 30% in cash immediately and pay Tk. 9,30,000 at the end of the each year for 10 years.

(iii) Pay 50% in cash and take a loan of Tk. 45,00,000 from AB Bank Ltd. @ 15% p.a. for 20 years. Assume the opportunity cost 12%. Which offer should you accept?

Solution : ৪৫ পৃষ্ঠার **Problem-44** নং প্রশ্নের উত্তর দেখুন।

Chapter 03 Capital Budgeting

(মূলধন বাজেটিং)

Part-A (Brief Questions):

001. What is capital budgeting? (মূলধন বাজেটিং কি?

[N.U. BBA (Hon's) Marketing- 2018, 2021, 2023]

Ans : Capital budgeting is the process of identifying, analyzing and selecting investment projects whose returns (cash flows) are expected to extend beyond one year.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

002. What is business risk?

[N.U. (Marketing)2017, 2021, 2023]

Ans. Business risk refers to the risk associated with the firm's operations its impact is shown in the varibility of firms operating income (EBIT).

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

oo003. What is risk? (ঝুঁকি কি?) ৯৯%

[Test Exam : Eden, Kobi Nozzrul, Rajandro, Jasseor City College-N.U. 2014, (Marketing) 2016]

Ans : Risk is the financial loss or more formally, the variability of returns associated with a given assets.

ooo04. What is risk premium? (ঝুঁকি প্রিমিয়াম কি?)

[Test Exam : Eden & Suhrawady College]

Ans : The excess return required from an investment in a risky assets over that required from a risk free investment is known as the risk premium.

ooo05. What is return? (আয় কি?)

[Test Exam : Eden, Habibulla College]

Ans : The total gain or loss experienced on an investment over a given period of time.

006. What is uncertainty?

(অনিশ্চয়তা কি?) ৯৯% *[N.U. 2019]*

Ans : Uncertainty refers to the outcomes of given event which are too unsure to be assigned probabilities.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

oo007. What is certainty equivalution (CE)? (নিশ্চয়তা সমমূল্যতা কি?)

Ans : The amount of cash an individual would require with certainty at a point in time to make them indifference between receiving that certain amount and receiving a risky expected amount at the same time.

✪✪✪08. **What is risk-adjusted discount rate (RADR)? (ঝুঁকি সমন্বিত বাট্টা হার কি?)** 99%

Or, What is RADR?

[টেস্ট পরীক্ষা– ঢাকা ও হাবীবুল্লাহ কলেজ; N.U. 2014]

Ans : The risk adjusted discount rate is the rate of return that must be earned on a given project to compensate the firm's owners adequately-that is, to maintain or improve the firm share price.

✪✪✪09. **What is sensitivity analysis? (সংবেদনশীলতা বিশ্লেষণ কি?)** 99%

[Test Exam : Dhaka, Adward & Titumir College;N.U. 2013]

Ans : Sensitivity analysis indicates exactly how much NPV will change in response to given change in an input variable, other things held constant.

✪✪✪(10) **What is cash conversion cycle?** 99%

[D.U. (Finance 3rd)-2016; N.U. Old 2012; 4th Acc-2015, 2018, Mgt-2018, Finance-2019; Acc 3rd-2018, 2023]

Ans. The flow of cash from purchase of inventory to production, from production to sell of goods and collection of receivables and again purchase of raw material can be termed as the cash conversion cycle.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। –সম্পাদক]

✪✪✪11. **What is capital asset pricing?**

[D.U. (Affiliated) 2016]

Ans. Capital asset pricing helps in calculating the expected rate of return from a share of equivalent risk in the capital market.

✪✪ 12. **What is an opportunity cost? (সুযোগ ব্যয় কি?)**

[টেস্ট পরীক্ষা– হাবীবুল্লাহ, বিএম, বরিশাল কলেজ]

Ans : What is lost by not taking the next best investment alternative is called opportunity cost.

✪✪ 13. **What is sunk cost? (নিমজ্জিত ব্যয় কি?)**

[টেস্ট পরীক্ষা– হাবীবুল্লাহ, বিএম, বরিশাল কলেজ]

Ans : A sunk cost is one that has already occurred and is not affected by the capital project decision.

✪✪✪14. **What is pay back period? (পরিশোধ সময় কি?)** 99% *[N.U. (Marketing) 2017, 2020, 2022]*

Or, What is meant by PBP?

Ans. The payback period is the time required for the firm to recover its initial investment in a project, as calculated from cash flows.

[রে স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। –সম্পাদক]

✪✪✪(15) **What is Internal Rate of Return (IRR)? (অভ্যন্তরীণ উপার্জন হার কি?)** 99%

[টেস্ট পরীক্ষা– রাজেন্দ্র, তোলারাম ও যশোর সিটি কলেজ; N.U. 2012, 2016, 2021, 2023]

Or, What is IRR.

Ans : The internal rate of return is the discount rate that equates the sum of the present values of series cash inflows with the initial investment.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। –সম্পাদক]

✪✪✪16. **What is average rate of return? (গড় উপার্জন হার কি?)**

Ans : The average rate of return is capital budgeting method that compares average after tax net cash inflow to average investment.

✪✪ 17. **What is net present value? (নিট বর্তমান মূল্য কি?)** *[Test Exam : Azam khan sout College]*

Ans : Net Present Value is the present value of an investment project's net cash flows minus project's initial cash outflows.

✪✪✪18. **What is the profitability index(PI)?** 99% **(মুনাফার্জন সূচক কি?)**

[টেস্ট পরীক্ষা– তোলারাম কলেজ; N.U. 2019]

Ans : Profitability index be defined as the ratio which is obtained by dividing the sum of present value of future cash inflows by the present value of the cash outlay.

19. What is capital rationing? (মূলধন বরাদ্দকরণ কি?) [২০২৪ সালে যা এসেছিল] [N.U. 2016, 2019, 2024]

Ans : The capital rationing refers to the choice of investment proposals under financial constraints in terms of a given size of capital.

✪✪ 20. **What is decision tree approach?**

Ans : A decision tree is a graphic presentation of the relationship of the present decision with future events and decision.

✪✪✪21. **What is scenario analysis? (দৃশ্যপট বা সিনারিও বিশ্লেষণ কি?)**

[Test Exam : Suhrawardy, Edan College, N.U. 2015]

Ans : Scenario analysis is a risk analysis technique in which the best and worst case NPV are compared with the projects expected NPV.

✪✪✪(22) **What is floating cost?** 99% *[N.U. 2021, 2023]*

Ans. The cost that is occurred in issuing securities is called floating cost.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। –সম্পাদক]

Part- B & C (Short & Broad Questions) :

○○○ 01. Defined capital budgeting. [Ans মার্কেটিং ইজিপ্লেস (২য় খণ্ড) বইয়ের 51 Page 3.01 No.]

[টেস্ট পরীক্ষা- বিএম, এমএম ও মহসিন কলেজ]

○○○ (02) **Show the difference between risk and uncertainty? 99% [Ans : 55 Page 3.15 No.]**

[N.U. (Hon's) 2009, 2011; 2018 (Accouting-3rd) 2018 (Managment-4th); D.U. 2016 (Accouting-3rd)]

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

○○ 03. Distinguish between NPV & IRR. **[Ans : 54 Page 3.10 No.]**

○○○ 04. **Distinguish between certainty equivalent approach and risk adjusted discount rate approach. 99%**
Or, Difference between CE and RADR. **[Ans : 56 Page 3.18 No.]** ***[N.U 2014; D.U. 2016 (Accouting-3rd)]***

○○ 05. What is meant by risk? **[Ans : 57 Page 3.19 No.]** ***[N.U. BBS(Hon's) 2006, 2008, 2011]***

○○○ 06. **(a) Discuss in brief the techniques of capital budgeting. 99% [Ans : 53 Page 3.05 No.]**
[টেস্ট পরীক্ষা- কবি নজরুল, হাবিবুল্লাহ, বরিশাল, এডওয়ার্ড ও যশোর সিটি কলেজ; N.U. 2013, 2015; 2016 (Accouting-3rd);D.U. 2017 (Accouting-3rd)]

(b) What are the techniques of adjusting risk in capital budgeting? Explain. 99%
[Ans : 56 Page 3.17 No.] *[টেস্ট পরীক্ষা- বরিশাল ও হাবিবুল্লাহ কলেজ; N.U -2012, 2014]*
Or, Explain the CE Approach of adjusting risk is capital budgeting.

○○ 07. Which is the best method of evaluation of project out of net present value (NPV) and internal rate of return (IRR)? ***[N.U. BBS(Hon's) 2020]***

○○○ (08) **Discuss the importance of capital budgeting. 99% [Ans : 52 Page 3.03 No.]**
[Test Exam : Dhaka, Titumir, Borishal College; N.U. BBS(Hon's) 2006, 2008, 2011; 2017 (Accouting); (Marketing-3rd) 2021, 2023]
Or, Importance/Advantages of capital budget.

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

Help Line ***Dear Students, In above Suggestions Part- B and C : 02, 06 & 08 No. Questions in are most important for the Examinee of 2025. ..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

○○ **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old) 2015 No– 03**

Canvass Ltd. is considering two mutually exclusive project. Project A and project B. A requires an initial outlay of Tk. 20,000 and the expected cash inflows of Tk. 5,000 each of the next 5 years. Project B requires an initial outlay of Tk. 25,000 and has an expected cash inflows of Tk. 6,500 for each of the next 5 years. Use pay back period method which project do you recommend for the company.

Solution : মার্কেটিং ইজিপ্লেস (২য় খণ্ড) বইয়ের ৫৯ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) Old 2012; 2nd Marketing 2015; 3rd Marketing 2021 No–05; 2023 No–09**

Cost of equipment Tk. 5,00,000. Expected life 5 years, salvage value at the end of life Tk. 50,000, annual CFBT Tk. 1,80,000, corporate tax rate 40%. Calculate PBP and comment on the result.

Solution : ৬২ পৃষ্ঠার **Problem-07** নং প্রশ্নের উত্তর দেখুন।

○○○ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

২০২৪ সালের পরীক্ষায় যা এসেছিল

⊗⊗ **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2024 No–08**

Cash inflow after tax (CFAT) of Project-A of BD company are :

Year	0	1	2	3	4	5
Project-A	(11,111)	4,444	3,333	9,999	5,555	9,789

Calculate payback period (PBP).

Solution : ৫৩৯ পৃষ্ঠার **Question No-8** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Acc. (1st Year) 2019**

Initial cash outlay of a project is Tk. 50,000. The expected annual equal cash inflows over the next 5 years life are Tk. 15,000. Calculate pay back period and pay back reciprocal of the project where corporate tax rate is 35%.

Solution : ৬২ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2016 No–11**

Green Ltd. is considering two mutually exclusive projects. The expected cash flows of each project are given below :

Year	Project- A	Certainty equivalent	Project-B	Certainty equivalent
0	(3,00,000)	1.0	(3,00,000)	1.0
1	1,00,000	0.95	2,00,000	0.90
2	2,00,000	0.90	2,00,000	0.80
3	2,00,000	0.85	2,00,000	0.70
4	3,00,000	0.80	3,00,000	0.60
5	3,00,000	0.75	4,00,000	0.50

The company has decided to evaluate these projects using certainty equivalent method. Company's normal required rate of return is 15% and after tax risk free rate of return is 8%.

Which project should be selected?

Solution : ৬৩ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Finance) 2019 No–8**

Bata Shoe Ltd. is evaluating a project with a beta of 1.50. The project has expected CFAT of Tk. 5,000 per year for four years, with an initial investment of Tk. 12,000. The risk-free rate is 8% and the market risk premium is 8%. You are required to determine whether the project should be accepted or rejected.

Solution : ৬৪ পৃষ্ঠার **Problem-10** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Accounting) 2019 No– 11**

ABC Ltd. has constructed a table, shown below, that gives expected cash inflows and certainty equivalent factors for these cash inflows. These measures are for a new machine with a five year life that requires an initial investment of Tk. 9,50,000. The firm has a 15% cost of capital and the risk free rate is 10% :

Year	CFAT Taka	Certainty equivalent factors
1	4,00,000	1.0
2	4,00,000	0.8
3	4,00,000	0.6
4	4,00,000	0.6
5	4,00,000	0.2

Required :

(a) What is the net present value (unadjusted for risk)?

(b) What is the certainty equivalent net present value?

Solution : ৬৫ পৃষ্ঠার **Problem-12** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old) 2013 No– 11**

Suppose there is a project which involves initial cost of Tk. 20,000 (cost at t = 0). It is expected to generate net cash flows during the first 2 years with the probability as shown in the following table:-

Year-1		Year-2	
Probability	**Net cash-flow**	**Probability**	**Net cash-flow**
0.10	6,000	0.10	4,000
0.25	8,000	0.25	6,000
0.30	10,000	0.30	8,000
0.25	12,000	0.25	10,000
0.10	14,000	0.10	12,000

(i) Calculate expected value of net cash flows for different periods.
(ii) Calculate the standard deviation of probable net cash-flows for different periods.
(iii) Calculate NPV if discount rate is 11%
(iv) Calculate the standard deviation of NPV.

Solution : ৬৯ পৃষ্ঠার **Problem-16** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old) 2012 No–10; 4th Year Management 2020 No–11**

Preynta Computers/Abrar Enterprize is considering investment in one of the mutually exclusive projects A and B. The firm's cost of capital (k) is 15% and the risk free rate (R_F) is 10%. The firm has gathered the following cash flow and risk index data for each project :

	Project	
	A	B
Initial investment	Tk. 15,000	Tk. 11,000
Year	Cash inflows	
	Taka	**Taka**
1	6,000	6,000
2	6,000	4,000
3.	6,000	5,000
4	6,000	2,000
Risk index	1.80	1.00

Required :
(a) Find the NPV of each project using the firms cost of capital.
(b) Determine the RADR using CAPM.
(c) Use the RADR for each project to determine its risk-adjusted NPV. Which project is preferable in this situation?.

Solution : ৭০ পৃষ্ঠার **Problem-17** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old) 2015 No–11**

Star Restaurant Ltd. is considering two mutually exclusive investment proposals. Their expected cash flow streams are given below :-

Project	NCO	CFBT per year	Life	Tax
A	15,000	5,000	5	40%
B	18,000	5,500	6	40%

After computing PBP the company compares RADR as follows :-

Less than one year	= 5%
1 to 3 years	= 8.5%
3 to 5 years	= 12.50%
Above 5 years	= 15%

Required : Find out the profitable project for the restaurant According to NPV.

Solution : ৭২ পৃষ্ঠার **Problem-19** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-11** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Accounting) 2020 No–11**

Tahmid Company Ltd. is considering an investment project with expected life of 5 years. The project requires an initial cash out flow of Tk. 1,00,000. The expected cash flows, certainty equivalent co-efficient and present value of Tk. 1@ 10% are as follows :

Year	CFAT	Certainty equivalent co-efficient	Present value @ 10%
1	20,000	0.95	0.909
2	60,000	0.80	0.826
3	40,000	0.70	0.751
4	30,000	0.60	0.683
5	15,000	0.50	0.621

The risk free rate of return is 10%. Compute the NPV and IRR. Should the project be accepted or not?

Solution : ৭৩ পৃষ্ঠার **Problem-20** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem–12** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2019 No–13**

From the following data of machine-A; you are required to calculate IRR :

		Tk.
	Machine cost	60,000
Annual estimated income after depreciation and income tax		
Year:	1	6,300
	2	5,400
	3	7,300
	4	10,000
	5	12,000
		41,000

Estimated life	5 years
Estimated salvage value	3,000
Average income tax rate	40%

Depreciation has been charged on straight line basis. Cost of capital is 12%.

Solution : ৭৬ পৃষ্ঠার **Problem-23** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-13** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Marketing 2017 No–12; 2023 No–13**

A chemical company is considering a project which requires an investment of Tk. 4,00,000. The estimates salvage value is zero. Tax rate is 50%. The company uses straight line depreciation and the proposed project has cash flows before tax (CFBT) as follows :

Year	CFBT
1	1,00,000
2	1,00,000
3	1,50,000
4	1,50,000
5	2,50,000

Determine :

(i) Payback period

(ii) Average rate of return

(iii) Net present value @ 15% PV factor.

Solution : ৭৭ পৃষ্ঠার **Problem-24** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem–14** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2021 No–16**

ABC company is considering a project which requires an investment of Tk. 4,00,000. The estimated salvage value is zero. Tax rate is 50% and cost of capital is 12%. The company uses straight line depreciation and the proposed project cash flows before tax (CFBT) are as follows :

Year	CFBT
1	Tk. 1,00,000
2	Tk. 1,00,000
3	Tk. 1,50,000
4	Tk. 1,50,000
5	Tk. 2,50,000

Required : (a) NPV (b) IRR.

Solution : ৭৮ পৃষ্ঠার **Problem-25** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-]

XX **Problem-15** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2016 No–12; 2022 No–11; 2024 No–11**

Dema Ltd./Padma Ltd is considering a proposal to install a new machine. The project will cost Tk. 1,00,000 and its expected life is 5 years. The estimated cash inflows before tax (CFBT) of the project as follows :

Year	CFBT (Tk.)
1	25,000
2	30,000
3	35,000
4	40,000
5	45,000

Requirements :

(a) PBP; (b) ARR; (c) NPV at 10%; (d) IRR; (e) Surplus life.

Solution : ৮১ পৃষ্ঠার **Problem-28** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-16** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Accounting (1st Year) 2016**

Talukder Enterprise is considering a new product line. The details of the investment proposals are as follows :

Initial cash outlay	Tk. 1,00,000
Expected life	5 years
Salvage value	Tk. 10,000
Working capital	Tk. 20,000

Cash flow before depreciation and taxes :

Years	CFBT (Tk.)
1	25,000
2	30,000
3	32,000
4	35,000
5	40,000

The project cost of capital is 10% and tax rate is 45%. Depreciation will be on sraight-line basis. You are required to calculate :

(i) Pay back period (PBP);
(ii) Average rate of return (ARR);
(iii) Return on investment (ROI).

Solution : ৮৪ পৃষ্ঠার **Problem-30** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-17** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Accounting (1st Year) 2019**

The cash flow for projects A and B are shown below. Each project has a cost of Tk. 2,50,000. The company's cost of capital is 10%

Year	Project -A	Project -B
1	Tk. 60,000	Tk. 65,000
2	90,000	65,000
3	1,00,000	65,000
4	1,10,000	65,000

(i) Should A or B accepted if they are independent project?

(ii) Which project should be accepted if they are mutually exclusive?

Solution : ৮৭ পৃষ্ঠার **Problem-33** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-18** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Finance (1st Year) 2020 No-14(b)**

AK Ltd. is considering two mutually exclusive projects. The firm, which has cost of capital is 11%. The following is the cash flows :

	Project	
	X	Y
Initial investment	Tk. 1,35,000	Tk. 92,000
Year	Net cash inflow (NCB)	
1	Tk. 40,000	Tk. 55,000
2	Tk. 28,000	Tk. 35,000
3	Tk. 45,000	Tk. 20,000
4	Tk. 35,000	Tk. 18,000
5	Tk. 45,000	Tk. 12,000

You are required to calculate IRR of each project and assess it acceptability.

Solution : ৮৮ পৃষ্ঠার **Problem-34** নং প্রশ্নের উত্তর দেখুন।

Chapter 04 Financial Statement Analysis
(আর্থিক বিবরণী বিশ্লেষণ)

Part-A (Brief Questions) :

০০০01. **What is financial statement? (আর্থিক বিবরণী কী?)** ৯৯%

Ans : Financial statement is a formal record of the financial activity and position of a business, person or firm.

Bank (02) **What is ratio analysis?** ৯৯%

[N.U. (Marketing)- 2019, 2023]

Ans. Ratio analysis is the process of determining and interpreting numerical relationships based on financial statements. Ratio analysis is used to evaluate various aspects of a company's operating and financial performance such as its efficiency, liquidity, profitability and solvency.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

০০ 03. **Write down the types of financial analysis (আর্থিক বিশ্লেষণের প্রকারভেদ লিখ।)**

Ans : (i) External, (ii) Internal, (iii) Vertical, (iv) Horizontal.

০০০04. **How many types of financial statement users? (আর্থিক বিবরণী ব্যবহার কারী কত প্রকার?)**

Ans : Two types : (i) Internal and (ii) External.

০০ 05. **Write down the efficiency ratio. (Efficiency ratio-গুলো লিখ।)**

Ans : Turnover ratio, expenses ratio, activity ratio.

০০০06. **What are the ratio under liquidity ratio? (তারল্য অনুপাতের অন্তর্ভুক্ত অনুপাতগুলো কী কী?)**

Ans : Net Working capital ratio, current ratio, acid test, super quick ratio.

○○ 07. **What are the capital structure ratios? (মূলধন কাঠামো অনুপাতগুলো কী কী?)**

Ans : Debt equity ratio, debt capital ratio, total assets ratio, capital gearing ratio.

○○ 08. **What are the ratio under summary of ratio? (অনুপাতসমূহের সারমর্ম এর মধ্যে কোন অনুপাতগুলো অন্তর্গত?)**

Ans : Liquidity, activity, debt, profitability and market ratio.

○○○09. **What is the formula of current ratio? (চলতি অনুপাতের সূত্র কী?)**

Ans : Current ratio = $\frac{\text{Current assets}}{\text{Current liabilities}}$

○○ 10. **What is the standard value of current ratio? (চলতি অনুপাতের আদর্শমান কত?)**

Ans : The standard value of current ratio is 2 : 1

○○○ 11 **Write down the formula of quick ratio. (তড়িৎ অনুপাতের সূত্রটি লিখি।)** 99%

[N.U. (Marketing)- 2018]

Ans : Quick ratio = $\frac{\text{Quick assets}}{\text{Current liabilities}}$

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○○12. **What is the standard value of quick ratio? (তড়িৎ অনুপাতের আদর্শমান কত?)**

Ans : The standard value of quick ratio is 1 : 1

○○ 13. **What is the standard value of creditors turnover? (পাওনাদর আবর্তনের আদর্শমান কত?)**

Ans : The standard value of creditors turnover is 6 and above times.

○○○14. **What is the formula of debtors turnover? (দেনাদার আবর্তনের সূত্রটি কী?)** 99%

Ans : Debtors turnover = $\frac{\text{Net credit sales}}{\text{Average debtors (A/R)}}$

○○○15. **Write down the standard value of debtors turnover. (দেনাদার আবর্তনের আদর্শমান লিখ।)**

Ans : The standard value of debtors turnover is 4 to 6 times.

○○ 16. **What is the elaboration of FASB? (FASB-এর পূর্ণরূপ কী?)**

Ans : Financial Accounting Standard Board.

○○○17. **What is P/E ratio? (P/E অনুপাত কী?)**

Ans : P/E ratio = $\frac{\text{Market price per share}}{\text{Earning per share}}$

○○○18. **Write down the formula of retention ratio. (সংরক্ষণ অনুপাতের সূত্র লিখ।)** 99%

[N.U. (Marketing)- 2016]

Ans : Retention ratio = 100% – D/P ratio.

○○ 19. **Write down the formula of gross profit ratio? (মোট মুনাফার সূত্র লিখ।)**

Ans : Gross profit ratio = $\frac{\text{Gross profit}}{\text{Sales}}$

20. *What is current ratio? [২০২৪ সালে যা এসেছিল]*

[N.U. (Marketing)- 2020, 2024]

Ans. The current ratio is a liquidity ratio that measures whether a firm has enough resources to meet its short-term obligations.

Part- B & C (Short & Broad Questions) :

○○○ 01. Who is financial manager? *[টেস্ট পরীক্ষা-মিরপুর, এম এম, সাদত কলেজ]*

[Ans : প্রাক্টিক্যাল মার্কেটিং হ্যান্ডবুক (২য় খণ্ড) বইয়ের 93 Page 4.01 No.]

○○ 02. What are the features of financial statement? **[Ans : 94 Page 4.02 No.]** *[টেস্ট পরীক্ষা-মিরপুর ইউনিভার্সিটি, এম এম কলেজ]*

○○○ 03. Who are the user's of financial statement? **[Ans : 94 Page 4.05 No.]**

○○○ 04. **What is ratio and ratio analysis?** 99% **[Ans : 94+96 Page 4.06+4.13 No.]**

○○○ 05. **(a) What is financial statement?** 99% **[Ans : 95 Page 4.08 No.]** *[টেস্ট পরীক্ষা-মিরপুর, সাদত কলেজ]*

(b) What are the objectives of financial statement? 99% **[Ans : 95 Page 4.09 No.]**

○○○ 06 **Discuss the cautions in using ratio analysis.** 99% **[Ans : 95 Page 4.10 No.]** *[N.U. (Marketing-3rd) 2016]*

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

○○○ 07: What are the techniques or tools of financial analysis? **[Ans : 94 Page 4.04 No.]**

Help Line ***Dear Students, In above Suggestions Part- B and C : 05 & 06 No. Question in are most important for the Examinee of 2025.*** ***..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

০০ **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Marketing 2018 No– 04**

The following data are taken from the financial statement of Morium Company :

	2013	2012
Accounts receivable (net) end of year	Tk. 5,50,000	Tk. 5,20,000
Net sales on account	Tk. 39,60,000	Tk. 31,00,000

Terms for all sales are 1/10 n/60

Compute for each year (l) The receivable turnover and (2) the average collection period. At the end of 2011 accounts receivable (net) was Tk. 4,80,000.

Solution : মার্কেটিং সাজেশন্স (২য় খণ্ড) বইয়ের ১০০ পৃষ্ঠার **Problem-03** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem–02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2019 No–07; 2022 No–06**

Jabber Ltd./Troll Ltd. has total sales of the year Tk. 6,90,000, cost of goods sold Tk. 6,55,000, Accounts receivable Tk. 70,000, Inventory Tk. 2,00,000.

Calculate the following ratios :

(i) Inventory turnover ratio

(ii) Accounts receivable turnover

(iii) Average collection period

Solution : ১০০ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

০০০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

০০০ **Problem–03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2021 No–06**

From the following information calculate (i) Current ratio and (ii) Acid test ratio :

Assets	Tk.	Liabilities	Tk.
Cash	20,000	Accounts payable	20,000
Accounts receivables	50,000	Expense payable	30,000
Inventory	70,000	Long-term loan	1,25,000
Prepaid expense	20,000	Capital	60,000
Fixed assets	75,000		
Total	2,35,000		2,35,000

Solution : ১০১ পৃষ্ঠার **Problem-04** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old) 2012 No–07; Marketing -2018 No–13; 2020 No–13; 2023 No–14**

Aurni Fashion House/Pran RFL/ABC company gives you the following information :

Accounts titles	Tk.
Merchandise inventory	1,20,000
Notes receivable	30,000
Cash in hand	45,000
Notes Payable	20,000
Account receivable	60,000
Cash at bank	90,000

Accounts payable	1,00,000
Accrued expenses	10,000
Unearned revenue	5,000

You are required to calculate :

(i) Gross working capital,
(ii) Net working capital,
(iii) Current ratio,
(iv) Acid test ratio.

Solution : ১০২ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

২০২৪ সালের পরীক্ষায় যা এসেছিল-

Problem-05 ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2024 No–12**

Jawwad Company has the following current assets and liabilities :

Accounts titles	Taka
Inventory	1,70,000
Cash	22,000
Accounts receivable	45,000
Bank overdraft	20,000
Accounts payable	45,000
Prepaid expense	12,000
Accrued salaries	22,000

During the year sales were Tk. 8,00,000 and purchase were Tk. 4,40,000. Determine the following ratios :

(a) Current ratio
(b) Acid test ratio
(c) Inventory turnover ratio
(d) Accounts receivables and payables turnover ratio.

Solution : ১০৩ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

Problem-06 ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2017 No–13; 2022 No–13**

The following balance sheet have been taken from the books of Muhsinat and Co./Tareque and Co.:—

Liabilities	Taka	Assets	Taka
Share capital	1,00,000	Fixed assets	1,19,000
Retained earnings	30,000	Inventories	1,20,000
Debenture	1,25,000	Debtors/Accounts receivable	90,000
Creditors	75,000	Cashable securities	8,000
Expenses payable	15,000	Prepaid expenses	10,000
Income tax payable	5,000	Cash	3,000
	3,50,000		3,50,000

You are required to compute :

(i) Working capital. (ii) Current ratio, (iii) Quick ratio, (iv) Proprietary ratio.

Solution : ১০৪ পৃষ্ঠার **Problem-07** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল]

⊗⊗ **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2017 No– 07; 2024 No–05**

A Company has the following information :

Accounts titles	Taka
Accounts receivable	40,000
Cash	10,000
Bank overdraft	10,000
Inventory	1,00,000
Accounts payable	40,000

Calculate the following ratios and comment on the solvency position and working capital :

(i) Current ratio

(ii) Acid test ratio

(iii) Working capital ratio.

Solution : ১০৬ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem–08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2020 No–15**

Assume that a firm has owner's equity of Tk. 1,00,000. The ratios for the firm are :

Current debt to total debt	0.40
Total debt to owner's equity	0.60
Fixed assets to owner's equity	0.60
Total assets turnover	2 times
Inventory turnover	8 times

Complete the following balance sheet, given the information above.

Liabilities and equities	Tk.	Asset	Tk.
Current debt	–	Cash	–
Long-terms debt	–	Inventory	–
Total debt	–	Total current assets	–
Owner's equity	–	Fixed assets	–
Total capital	–	Total assets	–

Solution : ১০৯ পৃষ্ঠার **Problem-12** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Marketing-2020 No–15; 4th (Acc.) 2015 No–16.**

Assume that a firm has owner's equity of Tk. 1,00,000. The ratios for the firm are :

Current debt to total debt	0.40
Total debt to owner's equity	0.60
Fixed assets to owner's equity	0.60
Total asset turnover	2 times
Inventory turnover	8 times

Complete the followine balance sheet, given the information above:

Liabilities and Equities	Taka	Assets	Taka
Current debt	.....	Cash	.....
Long-term debt	.....	Inventory	.....
Owner's equity	.....	Fixed assets	.....

Solution : ১১২ পৃষ্ঠার **Problem-14** নং প্রশ্নের উত্তর দেখুন।

Problem-10 ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Marketing 2017 No– 15**

Balance Sheet completion using ratios complete the 2016 balance sheet for Nasif Industries using the information that follows it :

Nasif Industries

Balance Sheet

December 31,2016

Assets	Taka	Liabilities	Taka
Cash	32,720	Accounts payable	1,20,000
Marketable securities	25,000	Notes payable	–.
Accounts receivable	–	Accrual expenses	20,000
Inventories	–	Total current liabilities	–
Total current assets	–	Long-term debt	–
Net fixed assets	–	Equity capital	6,00,000
Total assets		Total liabilities and OE	–

The following financial data for 2016 are also available :-

(1) Sales totaled Tk. 18,00,000.
(2) The gross profit margin was 25%.
(3) Inventory turnover ratio was 6 time.
(4) There are 365 days in the year.
(5) The average collection period was 40 days.
(6) The current ratio was 1.60 times.
(7) Total asset turnover ratio was 1.20 times.

Solution : ১১৩ পৃষ্ঠার **Problem-15** নং প্রশ্নের উত্তর দেখুন।

Chapter 05 Cost of Capital
(মূলধন ব্যয়)

Part-A (Brief Questions) :

01 **What is cost of capital? (মূলধন ব্যয় কী?)** ৯৯%

[N.U. 2022]

Ans. Cost of capital is minimum required rate of return.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। –সম্পাদক]

○○○02. What is meant by future cost? (ভবিষ্যৎ ব্যয় বলতে কী বুঝায়?)

Ans : Future cost means all the relevant cost related to an investment proposal.

○○ 03. Write down the defination of implicit cost. (অব্যক্ত ব্যয়-এর সংজ্ঞা লিখ।)

Ans : The expected rate of return of an alternative project that is rejected in order to invest in the present project is called the implicit cost.

○○○04. What is explicit cost? (ব্যক্ত ব্যয় কী?)

Ans : The cost that is directly paid to raise capital for a specific project is called an explicit or direct cost.

05 **What is WACC? (WACC কী?)** ৯৯%

[N.U. 2017, 2019, 2023]

Ans : Weighted average cost of capital (WACC) is the minimum rate of return a firm must earn on its assets to satisfy its investors.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। –সম্পাদক]

○○○06. What is full meaning WACC? (WACC এর পূর্ণরূপ কী?) ৯৯%

Ans : The full meaning of WACC is Weighted Average Cost of Capital.

○○○07. **What is meant by investing decision? (বিনিয়োগ সিদ্ধান্ত বলতে কী বুঝ?)**

Ans : The decision which is related to purchase of fixed assets and long-term investment is called investing decision.

○○○08. **What are the sources of cost of capital? (মূলধন ব্যয় হার নির্ধারণের উৎস কয়টি ও কী কী?)** 99%

[জা.বি. ২০১৬]

Ans : (i) cost of equity; (ii) Cost of debt; (iii) Cost of preferred stock; (iv) Cost of retained earnings.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○ 09. **What is cost of debt? (ঋণের ব্যয় কী?)**

Ans : The cost of long-term debt is the after-tax cost of raising funds through borrowing.

10. What is CAPM? [N.U. (Marketing 3rd)-2022, 2024]
Ans. The capital asset pricing model is a model that describes the relationship between the expected return and risk of investing in a security.

○○○ 11 **What is full meaning of CAPM?** 99%

[N.U. (Marketing)-2018]

Ans : CAPM = Capital Asset Pricing Model.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○ 12. **What is the elaboration of R_f? (R_F- এর পূর্ণ অর্থ কী?)**

Ans : Risk free rate of return.

○○○13. **What is breaking point? (ভগ্নবিন্দু কী?)** 99%

Ans : Breaking point is the limit of new financing till the weighted average cost of capital unchanged.

○○ 14. **What is NSV? (NSV কী?)**

Ans : Net sale value (NSV) is the amount of revenue a company retains from sales after adjusting for returns, allowances and discounts.

Part- B & C (Short & Broad Questions) :

○○○ 01 **What is cost of capital? What are the assumptions of cost of capital?**

[Ans : বিটাটেক্সন্স মার্কেটিং ইন্সপেক্স (২য় খণ্ড) বইয়ের 116+119 Page 5.01 + 5.12 No.]

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

○○ 02. What is weighted average cost of capital? [Ans : 116 Page 5.02 No.]

○○○ 03. **(a) What is dividend policy?** 99% [Ans : 117 Page 5.05 No.] *[N.U. 2012]*

(b) Discuss different types of dividend Policies. 99% [Ans : 118 Page 5.08 No.]

[N.U. 2012; 2018 (Accouting-3rd); D.U. 2016 (Accouting-3rd)]

○○ 04. What do you meant by opportunity cost? [Ans : 117 Page 5.06 No.]

○○○ 05 **Write down the assumption of explicit and implicit cost.** 99% [Ans : 119 Page 5.13 No.]

[N.U. 2020 (Marketing-3rd)]

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

Help Line ***Dear Students, In above Suggestions Part- B and C : 01 & 05 No. Questions in are most important for the Examinee of 2025.*** ***..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

○○○ **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2017 No–04; 2023 No–07**

Current market price is Tk. 300 per share, current dividend is Tk. 12 per share. Dividend per share is expected to increase at about 2% per share. Calculate the cost of equity share capital.

Solution : বিটাটেক্সন্স মার্কেটিং ইন্সপেক্স (২য় খণ্ড) বইয়ের ১২১ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

○○○ 99% [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

২০২৪ সালের পরীক্ষায় যা এসেছিল-

⊠⊠ **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2018 No– 06; 2022 No– 04; 2024 No–04**

XYZ Company/Walton Company/Jamuna Company paid Tk. 12 as dividend per share of common stock last year. Each share of Tk. 100 is currently selling at a premium of 5%. Dividend would grow by 5% per annum and flotation cost is 5%. Determine the cost of common stock.

Solution : ১২০ পৃষ্ঠার **Problem-01** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's)4th (Acc.) 2020 No–8**

Market price of common share of Moon Co. Ltd. is Tk. 110 per share and dividend per share is Tk. 10. Expected growth rate is 5%. Calculate cost of retained earning if personal tax rate is 40%.

Solution : ১২১ পৃষ্ঠার **Problem-03** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 4th (Acc.) 2018 No–7**

A 12% preference share of Tk. 1,000 each is issued at a discount of 7.50% and redeemable at a premium of 5% after 5 years. Flotation cost is 2% and dividend tax rate is 10%. Calculate the cost of preference share.

Solution : ১২২ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 4th (Acc.) 2016 No–7**

A firm has 80,000 ordinary shares outstanding. The current market price is Tk. 50 and the book value 35 per share. The firm's earning per share is Tk. 7.20 and dividend per share is Tk. 3.00. What is the growth rate assuming that the past performance will continue? Calculate the cost of equity capital.

Solution : ১২২ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Marketing-2017 No–16**

Webstar Company has compiled the information shown the following table :

Sources of capital	Book value (Tk.)	Market value (Tk.)	After tax cost
Long-term debt	40,00,000	38,40,000	6.0%
Preferred stock	40,000	60,000	13.0%
Common stock equity	10,60,000	30,00,000	17.0%
Total	51,00,000	69,00,000	

(i) Calculate the WACC using book value weights.
(ii) Calculate the WACC using market value weights.

Solution : ১২৩ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 2nd Marketing 2018 No-15**

Following is the data of X Ltd :

Details	Amount	Cost after tax
Common stock	20,00,000	12%
Preferred stock	2,00,000	9%
Retained earning	2,00,000	10%
Long-term debt	11,00,000	4%
	35,00,000	

In the next year the company would be needed an additional capital of Tk. 10,00,000. The present capital structure is the optimum capital structure.

Calculate : (a) How much amount should be financed from each source? (b) Calculate WACC.

Solution : ১২৪ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 1st Principle Acc. 2014 ; 3rd Year Marketing 2018 No– 14; 2022 No– 12**

Rosy/ABC Company was recently formed to manufacture a new product. It has the following capital structure :

Account titles	Taka
14% Debenture	12,00,000
12% Preferred stock	4,00,000
Common stock	16,00,000

The common stock sells for Tk. 50 and is expected to pay a Tk. 4 dividend this year. Which will grow at 10%. Corporate tax rate is 50%. Compute weighted average cast of capital.

Solution : ১২৫ পৃষ্ঠার **Problem-11** নং প্রশ্নের উত্তর দেখুন।

০০০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

০০ **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Year Marketing-2016 No–16**

The capital structure of Rupose Ltd. Company is as follows :—

	Taka
Common share	12,00,000
12% Preference share	5,00,000
8% Debenture	7,00,000
Retained earnings	8,00,000
	32,00,000

The common share of the company sell for Tk. 30 each. It is expected that the company will pay next year a dividend of Tk. 3.00 per share that will grow at 4% forever. The company's tax rate is 40%.
Calculate weighted average cost of capital.

Solution : ১২৬ পৃষ্ঠার **Problem-12** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

⊠⊠ **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2024 No–17**

The capital structure of Bangladesh Food Ltd. are as follows :

Accounts titles	Taka
Common stock (Tk. 100 per share)	20,00,000
14% Preferred stock (Tk. 1,000 per share)	10,00,000
15% Debentures (Tk. 2,500 per debenture)	15,00,000
Retained earnings	5,00,000
	50,00,000

Other information :

(i) Current market price of common stock in Tk. 150. Current year company declares a dividend of Tk. 15 per share. The company expects that this dividend will increase at 10% annual rate forever.

(ii) The company preferred stock is redeemable after 5 years at 10% premium, The flotation cost of preferred stock is 20% of sales price. Currently the preferred stock sales at Tk. 1,200.

(iii) The maturity period of debentures is 10 years. After 10 years debentures is redeemable at 12% premium. But currently sales at 5% discount. The flotation cost of debenture is 1.5% of its face value. The corporate tax rate 40% and personal tax rate 25%.

You are required to calculate the overall cost of capital.

Solution : ১২৭ পৃষ্ঠার **Problem-13** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem–11** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2020 No–16**

Raihan Company Ltd. has the following capital structure :

Particulars	Taka
Common stock (50,000 shares)	50,00,000
Retained earnings	30,00,000
12.5% preferred stock	20,00,000
14.5% debentures	20,00,000
Total capital	1,20,00,000

Additional information :

(i) The current market price of the share is Tk. 150 per share. Current dividend per share is Tk. 20 per share which is expected to grow at 9% per year and flotation cost of common stock is 4.5% of current market price.

(ii) The face value of preferred stock is Tk. 100 per share of which current market price is Tk. 120 and flotation cost is Tk. 8 per share. Maturity period is 5 years.

(iii) The par value of debenture is Tk. 100, which is sold for Tk. 90 and flotation cost is 5% of face value. Maturity period is 10 years.

(iv) The corporate tax rate is 40%.

Required : Find out the WACC using book value weight.

Solution : ১২৮ পৃষ্ঠার **Problem-14** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem–12** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2021 No–13**

The capital structure of XYZ company is as follows :

Particulars	Tk.
8% Debenture	1,00,000
12% Preference shares	50,000
Common shares	1,20,000
Retained earnings	30,000
Total	3,00,000

The common shares of the company sells for Tk. 30 each. It is expected that the company will pay next year a dividend of Tk. 2.5 per share that will grow at 4% forever. The market price of preference share is Tk. 95 (per value Tk. 100). A floatation cost of 5% of the market price would be incurred to the issue of all kinds of shares. The company's marginal tax rate is 40%. Compute the cost of capital of XYZ company.

Solution : ১৩১ পৃষ্ঠার **Problem-16** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-13** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) (Old) 1st 2008 ; 2nd Mgt. 2017 No-17**

Assuming corporate tax rate is 50% and personal income tax rate is 30%, find out the cost of the following sources :

(i) A 10 years 12% bond of Tk. 1,000 sold for Tk. 1,200 and 5% underwriting cost.

(ii) 15% debenture of Tk. 3,000 par value sold for Tk. 2,800.

(iii) 13% perpetual preferred stock of Tk. 100 par value sold for Tk. 120 with a flotation cost 5%.

(iv) 14% preferred stock of Tk. 100 par value ot be redeemed after 5 years, sold for Tk. 120 and flotation cost of Tk. 7 per share.

(v) Common stock selling at current market price of Tk. 175 and current dividend declared is Tk. 14 per share which is expected to grow at the rate of 6%, flotation cost being 5% of market price.

(vi) A common share of a company which does not believe in future external financing is selling for Tk. 150. the EPS is Tk. 25 and dividend payout ratio is 40%. the expected growth rate is 8%.

Required :

Calculate the WACC assuming the following weights are in the capital structure :

(i) 15%; (ii) 20%; (iii) 30%; (iv) 15%; (v) 15%; (vi) 5%.

Solution : ১৩৫ পৃষ্ঠার **Problem-19** নং প্রশ্নের উত্তর দেখুন।

Chapter 06 Working Capital Management
(চলতি/কার্যকরী মূলধন ব্যবস্থাপনা)

Part-A (Brief Questions) :

☑ Working Capital Management

01. What is working capital? (চলতি মূলধন কি?) [২০২৪ সালে যা এসেছিল]

[N.U. (Marketing)-2020, 2022, 2024]

Ans : Working capital simply refers to the capital which is required to finance current assets.

✪✪ 02. **What is gross working capital? (মোট চলতি মূলধন কি?)**

Ans : Gross working capital refers to the firm's investment in current assets.

✪✪✪03. **What is net working capital? (নিট চলতি মূলধন কি?)** *[Test Exam :Dhaka, Bangla, Rajendra College, N.U. 2015]*

Ans : Net working capital refers to the difference between current assets and current liabilities.

✪✪✪04 **What is operating cycle?** 99%

[টেস্ট পরীক্ষা- রাজেন্দ্র, মহসিন, এডওয়ার্ড ও যশোর সিটি কলেজ; N.U. 2014; (Marketing)-2018]

Ans. The time period between the requisition of inventory and the collection of cash from receivables is called operation cycle.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪ 05. **What is Permanent working capital? (স্থায়ী চলতি মূলধন কি?)**

Ans : Permanent working capital refers to the minimum amount of current assets that a business must maintain at all times to ensure uninterrupted operations.

✪✪✪06. **What is temporary working capital? (সাময়িক চলতি মূলধন কি?)**

Ans : The volume of current asset that raise and falls over and above the permanent as temporary working capital.

✪✪✪07. **What is working capital management? (চলতি মূলধন ব্যবস্থাপনা কি?)** 99%

[টেস্ট পরীক্ষা- তোলারাম, এডওয়ার্ড ও সাতক্ষীরা কলেজ; N.U. 2013]

Ans : Working capital management is the management of current assets, current liabilities and the relationship between them.

☑ Working Capital Financing

✪✪✪08. **What is short term finance? (স্বল্পমেয়াদি অর্থায়ন কী?)** 99%

Ans : The finance which is made for less than one year is called short term finance. It made to meet the daily activity of the organization.

✪✪✪09. **What is the ideal concept of short term finance? (স্বল্পমেয়াদি অর্থায়নের আদর্শ বিষয় কী?)**

Ans : When the finance is made for the current portion of the current asset is called the ideal concept of short term finance.

✪✪ 10. **What is a letter of credit? (প্রত্যয়পত্র কী?)**

Ans : A letter of credit is a letter issued by a bank to another bank to serve as a guarantee for payments made to specified person under specified conditions.

✪✪✪11. **What is spontaneous finance?**

Ans : Spontaneous finance arises from the normal course of business. Such as accounts payable.

12. What is trade credit? (ঋণের ব্যয় কী?) [২০২৪ সালে যা এসেছিল]

[N.U. (Marketing)-2017, 2020, 2022, 2024]

Ans : The credit which is created by purchasing of goods and services on credit is called trade credit.

✪✪✪13. **What is a marketable security?** 99%

[D.U. (Affiliated) BBA(Hon's) (Marketing)- 2016]

Ans. Marketable securities is short term securities which traded in money market and converted without any loss.

✪✪ 14. **What is meant by bank loan? (ব্যাংক ঋণ বলতে কী বুঝ?)**

Ans : When a bank makes an advance to a customer whether by over-draft or lone account is called bank credit.

✪✪✪ 15. **Why security is needed? (নিরাপত্তা কেন প্রয়োজন?)**

Ans : Security is needed to ensure the recovery of the loan.

✪✪✪16. **What is electronic box?**

[D.U. (Affiliated) BBA(Hon's) (Marketing)- 2016]

Ans. Box build is probably the most misunderstood aspect of contract electronics manufacturing.

✪✪✪17. **What is EIR?** 99% *[N.U. (Marketing)-2018]*

Ans. EIR = Effective interest rate is the annual interest rate that is actually paid or earned.

✪✪✪18. **Write the formula of calculating EIR in discounted method. (Discounted method-এ EIR নির্ণয়ের সূত্র লিখ।)** 99%

[N.U. (Marketing)-2018]

Ans : $EIR = \frac{Interest}{Loan - Interest} \times 100$

✪✪ 19. **How can we calculate the cost of commercial Paper? (বাণিজ্যিক পত্রের ব্যয় কীভাবে নির্ণয় করবে?**

Ans : $K_{cp} = \frac{FV - NSV}{NSV} \times \frac{360}{MP} \times 100$

✪✪✪20. **What is created in credit purchase? (বাকিতে ক্রয় কী সৃষ্টি করে?)**

Ans : Goodwill.

✪✪✪21 **What is effective annual interest rate?** 99%

[N.U. (Marketing)-2021]

Ans. An effective annual interest rate is the real return on a savings account or any interest paying investment when the effect of compounding over time are taken into accounts.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

Part- B & C (Short & Broad Questions) :

☑ Working Capital Management

✪✪✪ 01. What is working capital management?

[Ans মার্কেটিং ইন্টারেস্ট (২য় খণ্ড) বইয়ের 145 Page 6.15 No.]

[টেস্ট পরীক্ষা- ঢাকা, বিএম, এডওয়ার্ড ও যশোর সিটি কলেজ,N.U 2013; 2018 (Accouting-3rd) 2018 (Managment-4th) D.U. 2016 (Accounting)]

✪✪✪ 02 **Write down the external sources of working capital.** 99% [Ans : 145 Page 6.27 No.]

[টেস্ট পরীক্ষা : ইডেন, যশোর সিটি, আজিজুল হক ও এমএম কলেজ; [N.U. (Marketing-3rd) 2016]

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

✪✪✪ 03. Discuss the importance of working capital management of company. [Ans : 145 Page 6.18 No.]

[N.U. BBS (Hon's) 2007; 2018 (Accouting-3rd); D.U. 2016, 2017 (Accouting-3rd)]

✪✪✪ 04. Briefly discuss the objectives of working capital management. [Ans : 145 Page 6.16 No.]

[টেস্ট পরীক্ষা- আনন্দমোহন, কুষ্টিয়া, আজিজুল হক ও এমএম কলেজ] [N.U. 2015]

Or, What are the objectives of working capital management.

✪✪✪ 05. **Discuss different types of working capital in brief.** 99% [Ans : 146 Page 6.20 No.]

[N.U. 2014; 2017 (Finance-3rd)]

[২০২৪ সালের পরীক্ষায় যা এসেছিল–

13. Discuss the factors to be considered in determining working capital. [Ans : Page No. 143; Ques. No 6.11]

Or, Describe the determinants of working capital. *[N.U. BBS(Hon's) 2006, 2009, 2011; Marketing-2018, 2024]*

☑ Working Capital Financing

✪✪ 06. What is the cost of trade credit? [Ans : 151 Page 6.33 No.]

✪✪ 07. Who bears the cost of trade credit? [Ans : 151 Page 6.34 No.]

✪✪✪ 08 **Explain the aggressive approach of working capital financing.** 99% [Ans : 148 Page 6.23 No.]

[টেস্ট পরীক্ষা- বিএম ও কুমিল্লা ভিক্টো. কলেজ; N.U. 2015 (Marketing-3rd)]

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

Help Line ***Dear Students, In above Suggestions Part- B and C : 02 & 08 No. Questions in are most Important for the Examinee of 2025.*** ***..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

☑ Working Capital Management

ooo **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Marketing-2018 No–07; 2020 No–05**

The following are the information of Aurin Foods limited/Foods limited :

Inventory turnover 6 times

Average collection period 40 days

Average payment period 30 days

Assume 1 year = 360 days

Calculate :

(i) Operating cycle (OC).

(ii) Cash conversion cycle.

Solution : ব্যবস্থাপনা মার্কেটিং ইনভেস্টমেন্ট (২য় খণ্ড) বইয়ের ১৫৪ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

ooo (১০০%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

oo **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 4th (Acc.) 2018 No–06**

Your company has an average daily cash balance of Tk. 8,000. Total cash needed for the year is Tk. 5,80,000. The interest rate is 6% processing cost o Tk. 100 each transaction.

Calculate :

(a) Opportunity cost

(b) Trading cost

(c) Total cost

Solution : ১৫৫ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old) 2014 No–15(b)**

Jamuna company has annual sales of Tk. 5 million, cost of goods sold of 75% of sales and purchase that are 65% of cost of goods sold. Jamuna company has an average age of inventory of 50 days, an average collection period of 30 days and an average payment period of 20 days.

(i) Calculate operating cycle for Jamuna.

(ii) Calculate cash conversion cycle for Jamuna.

(iii) The resources Jamuna has invested in cash conversion cycle (assuming 360 day year.)

Solution : ১৫৭ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem–04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2020 No–07**

CPC Food Ltd. cash requirement Tk. 40,00,000 next year. The cost of fund is 10% per annum. The company has transaction cost of Tk. 100 per transaction.

Requirement : (i) Optimum cash balance; (ii) The total annual cost of demand for optimum cash balance.

Solution : ১৫৯ পৃষ্ঠার **Problem-11** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 4th (Accounting) 2017 No–06**

ABC Ltd. provides the following information :

(i) The standard deviation in daily cash balance is Tk. 10,000.

(ii) The minimum cash balance is to maintain Tk. 1,25,000.

(iii) The interest rate is 6% p.a.

(iv) The transaction cost for each time is Tk. 50.

Requirements :

(1) Return point (z);

(2) Target balance;

(3) Upper limit;

(4) Average balance.

Solution : ১৫৯ পৃষ্ঠার **Problem-12** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 4th (Acc.) 2013 No–11; 2016 No–13; 2020 No–12**

Your company has an average daily cash balance of Tk. 4,000. Total cash needed for the year is Tk. 4,80,000. The interest rate is 6% and processing cost of Tk. 100 each transaction.

Calculate :

(a) Opportunity cost.

(b) Trading cost.

(c) Total cost.

(d) Comment on your company's cash strategy.

Solution : ১৬০ পৃষ্ঠার **Problem-13** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Finance 2016 No–17**

Orion Shoes Ltd. has the following information about the cost of production :—

Particulars	Taka	Cost per unit
Raw material	120	
Direct labour	80	
Overhead	50	250
Profit		50
Selling price	Total cost	300

Additional information :-

(i) Stock of raw materials one month.

(ii) Work-in-process of material 20 days.

(iii) Stock of finished goods 1 month.

(iv) Credit allowed by creditors 40 days.

(v) Credit allowed to debtors 50 days.

(vi) Lag in payment of wages 12 days.

(vii) Lag in payment of overhead 30 days.

(viii) Cash sales 20% and credit sales 80%.

(ix) Cash in hand is of Tk. 1,00,000.

Requirement : Prepare a working capital statement and find out the net working capital assuming the annual production in unit is 1,50,000 units. Also assume that the work-in-process completion stage of 50% all item.

Solution : ১৬৩ পৃষ্ঠার **Problem-18** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old) 2014 No–16**

As a Management Accountant of Alfa Bangladesh Company, you are required to prepare a statement showing working capital requirement for 24,000 units of production in the next year. The following information is given below for this purpose.

(i) **Cost :**

Direct raw-materials cost per unit Tk. 20
Direct labor cost per unit Tk. 8
Overhead (100% of direct labor)
Profit 20% of selling price.

(ii) **Conversion period :**

(a) Raw-materials are expected to be in the stock for an average of 30 days before issue of production.
(b) Processing time is 15 days and finished goods are expected to remain in the store on an average for 45 days.
(c) Credit allowed to debtors 2 months; whereas the creditors allowed only 1 month.
(d) Time lag in payment of wages in 15 days.

Solution : ১৬৮ পৃষ্ঠার **Problem-21** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Finance 2017 No–17**

From the following information you working capital :

Particulars	Taka
Raw material	400 per unit
Direct labor	150 per unit
Overheads (excluding depreciation)	300 per unit
Total cost	850 per unit
Additional information :	
Selling price	1,000 per unit
Output	52,000 per units
Raw materials in store	average 4 weeks
Work-in-process (assumed 50% completion with full material consumption)	average 2 weeks
Finished goods in warehouse	average 4 weeks
Credit allowed by suppliers	average 4 weeks
Credit allowed to customers	average 8 weeks
Cash at bank is expected to be	Tk. 50,000
25% of purchases are made for cash.	

Assume that production is sustained at an even pace during the 52 weeks of the year. All sales are on credit basis.

Solution : ১৭০ পৃষ্ঠার **Problem-23** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 4th (Accounting) 2015 No–12; 2020 No–11**

From the following data of a company, estimate the amount of working capital needed to finance an activity level of 60%. The capacity of the concern to product 2,50,000 units per annum (p.a) :

(i) Expected selling price Tk. 10.00 per unit.
(ii) Cost of raw material Tk. 3.00 per unit.
(iii) Direct labour cost Tk. 2.50 per unit.
(iv) Annual overhead at 60% activity level (including depreciation Tk. 60,000) Tk. 3,00,000.
(v) Raw materials are in stock and average for one month.

(vi) Materials are in process on average for two months.
(vii) Finished goods are in stock an average for two months.
(viii) Credit allowed to debtors for three months and that received from suppliers of raw materials for one month.
(ix) Lag in payment of wages $\frac{1}{2}$ month and overhead one month.
(x) Cash in hand and at bank 10% of net working capital.
You may assume that production is carried on evenly throughout the year and overhead accrue similarly. One fourth of the output is sold against cash.

Solution : ১৭৩ পৃষ্ঠার **Problem-25** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-11** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old) 2013 No–15; 4th (Accounting) 2016 No–12**

ABS Ltd/Z and Z Co. Ltd. has been operating at 24,000 units of production per year with the following cost structure during 2015 :

Raw materials	Tk. 60 P.U
Direct labour (40% fixed)	Tk. 50 P.U
Factory overhead (80% fixed)	Tk. 40 P.U
Selling overhead	Tk. 10 P.U
Total	160 P.U
Add : Profit	40 P.U
Selling price	200 P.U

As on 31st December 2015, the company had the following balances :

Inventory of raw materials	Tk. 3,60,000
Inventory of WIP	Tk. 2,20,000
Inventory of finished goods	Tk. 7,20,000

It is estimated that the demand of its products will increase and the company is planning to increase production 50% in the next year. It is expected that because of increased volume of purchase, the supplier of raw materials will reduce direct materials cost by 10%. Credit period allowed to debtors 90 days and creditors allowed 2 months. Lag in payment of wages is 15 days and other expenses is one month.
You are required to estimate the working capital requirement for the 2016 assuming minimum cash balance requirement of Tk. 3 lakh.

Solution : ১৮১ পৃষ্ঠার **Problem-30** নং প্রশ্নের উত্তর দেখুন।

☑ Working Capital Financing

☑ Cost of Trade Credit

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

☒☒ **Problem-12** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2016 No–07; 2022 No–08; 2024 No–03**

ABC Company/Adib Company/Zabeer Company extended credit facility for Tk. 10,00,000 on terms of (i) 2/15, net 45 (ii) 2/20, net 90. Calculate which alternative is appropriate for the company.

Solution : ১৮৫ পৃষ্ঠার **Problem-32** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-13** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Finance (1st Year) 2018**

Calculate the EIR from the following situation's :
(a) Credit purchase Tk. 1,00,000, terms 2/12, Net 30 EOM.
(b) Credit purchase February 15, 2017 Tk. 50,000, terms 2/10, Net EOM.

Solution : ১৮৬ পৃষ্ঠার **Problem-33** নং প্রশ্নের উত্তর দেখুন।

☑ Cost of Commercial Paper

○○ **Problem-14** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 2nd Marketing 2017**

A commercial paper with face value of Tk. 10,000 each sold at Tk. 8,500 for 120 days. What will be the cost of commercial paper?

Solution : ১৮৮ পৃষ্ঠার **Problem-36** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem- 15** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 2nd Management 2020 No-08**

Raju Co. Ltd. has just issued Tk. 2,00,000 of commercial paper that has 3 months maturity period and sells Tk. 1,90,000. Calculate (i) APR (ii) EIR.

Solution : ১৮৮ পৃষ্ঠার **Problem-38** নং প্রশ্নের উত্তর দেখুন।

☑ A/R Factoring

○○○ **Problem-16** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Accounting (1st Year) 2019**

Haque Company's monthly credit sales is Tk. 1,50,000 and average collection period 72 days. The company is considering the factoring it's receivable. A factor has agreed to advance up to 75% of the face value at 15% interest p.a. The factor will also charge 2% commission. Factoring would save company's 0.5% bad debt loss and administration cost of Tk. 1,000 per month. Calculate net periodic cost.

Solution : ১৯০ পৃষ্ঠার **Problem-41** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল]

☒☒ **Problem-17** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2024 No–07**

Lotus Company has a revolving credit of Tk. 5,00,000 at an interest of 13% with Agrani Bank PLC. Average borrowing of the company was Tk. 3,00,000 for the year. A commitment fee of 2.5% is to be paid. Compute EIR?

Solution : ৫৩৮ পৃষ্ঠার **Question No-7** নং প্রশ্নের উত্তর দেখুন।

☑ Cost of Trade Credit

○○ **Problem-18** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 2nd Mgt 2015 No-14**

The credit terms for some alternatives are given below :

Alternative	Credit terms
X	2/10, net 60
Y	3/15, net 80
Z	2/20, net 70

(i) Determine the cost of foregoing cash discount for each alternative.
(ii) The firm has an another option to raise short-term bank loan @ 13% interest, Should the firm accept the cash discount offer or not?
(iii) What would be the decision in (ii) if accounts payable are stretched by 30 days.

Solution : ১৯০ পৃষ্ঠার **Problem-42** নং প্রশ্নের উত্তর দেখুন।

☑ Mixed Problem

○○ **Problem-19** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Finance (1st Year) 2020 No-11(b)**

Z Ltd. needs working capital of Tk. 10,00,000.
There are two alternatives :
(a) To forgo cash discount granted on the basis of 2/15 net 40
(b) To borrow from Sonali Bank Ltd. at 12.25% p.a. The bank requires 17.50% compensating balance on loan amount.
Which alternative is the best and why?

Solution : ১৯২ পৃষ্ঠার **Problem-43** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-20** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 2nd Management 2020 No-16**

Shapla Ltd. needs Tk. 40,00,000 to meet it's requirement and it has three alternatives :

(i) Trade credit: Buy 40,00,000 of raw materials 3/20, net 60 terms.

(ii) Bank loan : A bank will lend Tk. 40,00,000 at 11%, with 15% compensating balance interest to be paid in advance.

(iii) Commercial paper : Issue Tk. 40,00,000 for six months period and sales value Tk. 37,50,000. Flotation cost 2%.

Which alternative should the company choose and why?

Solution : ১৯৪ পৃষ্ঠার **Problem-46** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem–21** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2021 No–17**

Sara Ltd. requires working capital of Tk. 20 lakh. There are three alternatives :

(i) Trade credit : Terms of sale 2/20, net 90.

(ii) Bank loan : Trust bank is interested to grant Tk. 20,00,000 as a loan @ 10% per annum after reduction of 25% as compensating balance.

(iii) Commercial paper : To issue a commercial paper @ 10%. The issuing cost would be Tk. 20,000 in each quarter.

Which option the company should accept?

Solution : ১৯৭ পৃষ্ঠার **Problem-49** নং প্রশ্নের উত্তর দেখুন।

০০০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

০০০ **Problem–22** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2019 No–17**

Afroza Company needs Tk. 1,60,000 to meet working capital requirement. It has three alternative sources :

(i) The company can buy Tk. 2,00,000 of materials on terms of 3/30, net 90.

(ii) The company can take advance of Tk. 2,00,000 at 13% interest rate with 20% compensating balance requirement.

(iii) A factor will buy company's accounts receivable and will advance at 12% interest p. a. upto 80% of the face value. Annual credit sales Tk. 12,00,000 and average collection period is 60 days. The factor will charge 2% commission. It has been estimated that there will be administration cost of Tk. 1,000 per month and 1% bad debt loss.

Required : Which alternative should the company choose?

Solution : ১৯৯ পৃষ্ঠার **Problem-51** নং প্রশ্নের উত্তর দেখুন।

Chapter 7.1 Intermediate Term Financing

(অন্তর্বর্তীকালীন অর্থায়ন)

Part-A (Brief Questions):

০০০01. **What is intermediate term finance?** 99%
Ans : When the finance is made for less then 5 years and above 1 year it is called mid-term finance.

০ 02. **Write the name of four methods of mid-term finance.**
Ans :
(i) Bank term loan;
(ii) Leasing,
(iii) Revolving credit;
(iv) Insurance companies term loan.

০০০03. **What is revolving credit?**
Ans : A line of credit guaranteed to a borrow by a commercial bank is called revolving credit.

০০ 04. **What is Commitment Fee?**
Ans : Commitment fee is paid for the unused portion of loan

০০০05. **What is the elaboration of IMF?**
Ans : The elaboration of IMF is International Monetary Fund.

০০ 06. **What is the elaboration of BDBL?**
Ans : Bangladesh Development Bank Limited.

(07) **What is international lease?** 99%
[N.U. (Marketing)- 2016]
Ans. If the parties to a lease agreement are based in different countries, it's known as international lease.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

০০ 08. **What is the elaboration of BKB?**
Ans : Bangladesh Krishi Bank.

০০০09. **What is compensating balance?**
Ans : The certain portion of loan that can not be drawn by the borrower.

Part- B & C (Short & Broad Questions):

○○ 01. Define of mid-term financing. [Ans মার্কেটিং ইনসাইটস (২য় খণ্ড) বইয়ের 200 Page 7.1.01 No.]

○○○ 02. Describe-the characteristics of mid-term of financing. [Ans : 200 Page 7.1.02 No.]

○○○ 03. Who are the users of intermediate-term financing? 99% [Ans : 201 Page 7.1.04 No.]

○○○ (04) What are the different types of term loan? Discuss the uses of term loan. 99%

[Ans : 202 Page 7.1.07 No.] *[D.U. (Affiliated) BBA(Hon's) (Marketing)- 2016]*

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

Help Line *Dear Students, In above Suggestions Part- B and C : 04 No. Question in are most important for the Examinee of 2025.* ..... Editor

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

○○ **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 2nd Marketing 2019 No-14**

Bay Ltd. required Tk. 10,00,000 to finance its mid-term capital for production purpose. The AB Bank Ltd. agreed to give this loan at 15% half yearly rate of interest. The loan should be paid in next 5 years. You are required to prepare a loan amortization schedule under ballon method.

Solution : মার্কেটিং ইনসাইটস (২য় খণ্ড) বইয়ের ২০৩ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) Finance (1st Year) 2014 No-16; 2nd Management 2019 No-15**

Suppose that you borrow Tk. 5,00,000 from Uttara Bank Ltd. at 12% interest rate to be repaid over next 6 years; payments will be made in equal installments at the end of each period. How much should be paid in each installment? You are also reuired to show the repayment schedule. (Use Capital Recovery method).

Solution : ২০৪ পৃষ্ঠার **Problem-04** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 2nd Marketing 2016 No-14**

Habib company has taken a loan of Tk.1,00,000 from an insurance company for 4 years. The loan will be repaid over the next 4 years in equal installments. Interest rate is 10%. Prepare a loan repayment method under capital recovery method.

Solution : ২০৫ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-04** ***Year Question & Solution*** ⊙ **D.U. BBA (Hon's) 3rd Marketing 2016 No–11; N.U. BBA (Hons.) 3rd Marketing 2021 No–14; 2023 No–12**

Zaria Ltd./Mubira Ltd. needs Tk. 5,00,000 to finance its intermediate term capital requirement for remodeling of product processing equipments. Rupali Bank Ltd. has agreed to provide loan at 14% interest. The loan has to be paid in next 5 years. You are required to show the payment schedule under :

(a) Balloon payment method; (b) Capital recovery method.

Solution : ২০৭ পৃষ্ঠার **Problem-07** নং প্রশ্নের উত্তর দেখুন।

Chapter 7.2 Leasing (ইজারা)

Part-A (Brief Questions) :

01. What is lease financing? 99% (ইজারা অর্থায়ন কি?) *[Test Exam : Dhaka, Eden, Kobi Nazzrul Mohishin, Tularam, Azam khan, adward & uttara town College; N.U 2012, 2020, 2022]*

Ans : Lease financing is the process by which a firm can obtain the use of certain fixed assets for which it must make a series of contractual, periodic, tax deductible payments.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○○02. What is lease? *[N.U. 2014]*

Ans. Lease is a contract between a lessors, (the owner of the assets) and a lessee (the user of the assets).

○○ 03. What is financal lease? (আর্থিক ইজারা কি?)

Ans : Financial lease is a long-term lease than an operating lease that is non cancelable and obligates the lessee to make payments for the use of an asset over a predefined period of time.

04. What is operating lease? (পরিচালনা ইজারা কি?) 99%

[Test Exam : Dhaka, Suhraward, BM & Tularam College; N.U. 2013; (Marketing)- 2017, 2023]

Ans : Operating lease is a cancelable contractual arrangement whereby the lessee agrees to make periodic payments to the lessor.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○○05. What is direct lease? (প্রত্যক্ষ ইজারা কি?)

Ans : When the lease belongs to the owner of the assets and users of the assets with direct relationship it is called as direct lease.

06. What is sale and leaseback? (বিক্রয় ও ইজারা গ্রহণ কি?) 99%

[Test Exam : Edem, Habibulla, BM & Rajendra College, N.U. 2014; D.U. (Affiliated) 2016]

Ans. Under the sale and leaseback arrangement, the firm sells an asset that it owns and then leases to same asset back from the buyer.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○○07. What is leveraged lease? (লিভারেজ ইজারা কি?)

Ans : Leveraged lease is the same as the direct lease, except that a third party, the lender is involved in addition to the lessee and the lessor.

○○ 08. What term loans? (মেয়াদি ঋণ কি?)

Ans : Loans that have maturities of 1 to 10 years and are repaid in periodic installments over the life of the loan; usually secured by a chattel mortgage on equipment or a mortgage on real property .

○○○09. Who is lessor? (ইজারাদাতা কি?)

Ans : Lessor is the owner of the assets that are being leased. Lessors may be individual partnership, joint stock companies, corporation or financial institutions.

○○ 10. Who is lessee? (ইজারা গ্রহীতা কে?)

Ans : Lessee is the receiver of the service of the assets under a lease contract. Lease assets may be firms or companies.

Part-B & C (Short & Broad Questions) :

○○ 01. Define lease financing. **[Ans** মার্কেটিং ইনসাইটস (২য় খণ্ড) বইয়ের **208 Page,7.2.01 No.]**
Or, What is lease financing?

○○○ 02. Explain the impact of financial leasing on the cash-flow of a firm. **[Ans : 209 Page 7.2.04 No.]**
[টেস্ট পরীক্ষা- রাজেন্দ্র, যশোর সিটি ও শেরপুর কলেজ] N.U 2012; D.U. 2017 (Accouting-3rd)

○○○ 03. Discuss the advantages of lease financing. **[Ans : 210 Page 7.2.05 No.]**
[টেস্ট পরীক্ষা- বিএম, বরিশাল, তোলারাম কলেজ] [N.U. BBS(Hon's) 2008, 2010; 2018 (Accouting-3rd); D.U. 2016 (Accouting-3rd)]

○○○ 04. What is sale and leaseback? 99% [Ans : 210 Page 7.2.07 No.]

○○ 05. What is hire purchase? **[Ans : 210 Page 7.2.08 No.]** *[টেস্ট পরীক্ষা- ঢাকা, এসকে ও যশোর সিটি কলেজ]*

○○○ 06. Discuss different classification of lease. **[Ans : 209 Page 7.2.03 No.]**
Or, Narrate the tyres of lease financing. *[টেস্ট পরীক্ষা- শহীদ বুলবুল ও রাজশাহী কলেজ; N.U 2014]*

ooo 07. **Show the difference between lease financing and hire purchase.** 99% *[N.U Marketing-2020]*

[Ans : Page No-211 ; Ques. No- 7.2.11]

ooo (08) **Show the differences between operating lease and financial lease.** 99% **[Ans : 211 Page 7.2.10 No.]**

Or, What are the differences between operating lease and financial lease?

[টেস্ট পরীক্ষা- শের বোরহানুদ্দীন, বিএল, রাজবাড়ী ও রাজেন্দ্র কলেজ; N.U 2012, 2016 (Accouting-3rd); 2017 (Finance-3rd); BBS(Hon's) 2009, 2011, 2012, 2018, 2021 (Marketing-3rd)]

Or, Difference/Distinction/Distinguish between operating lease and financial or capital lease.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo (09) **Show the difference between lease and purchase.** 99% **[Ans : 213 Page 7.2.16 No.]**

Or, Distinguish between lease and purchase. *[টেস্ট পরীক্ষা- কবি নজরুল কলেজ] N.U 2014; Marketing-2018, 2023]*

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

Help Line ***Dear Students, In above Suggestions Part- B and C : 08 & 09 No. Questions in are most important for the Examinee of 2025.*** ***..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

oo **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 4th (Mgt) 2018 No–04**

From the following information, compute the annual lease payment (paid in advance) that a lessor will require-

(i) Purchase price of Tk. 2,50,000, interest rate 12%, 5 years lease period and no residual value.

(ii) Purchase price of Tk. 1,50,000, interest rate 8%, 8 years lease period and a near certain residual value of Tk. 20,000.

Solution : ব্যাচেলর মার্কেটিং ইন্টেলেক্স (২য় খণ্ড) বইয়ের ২১৭ পৃষ্ঠার **Problem-04** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2023 No–16**

Abul Khair Company is thinking to purchase a machine for Tk. 12,290. The company can borrow Tk. 12,290 at an interest rate of 10% to be repaid in 10 equal annual installment of Tk. 2,000 or it can buy the machine for an annual lease payment of Tk. 2,000. Under income tax act straight-line lease of depreciation is allowed. The company pays tax @ 50%. Which alternative should the company accept?

Solution : ২১৯ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old) 2012 No–14; Marketing 2016 No–15; 2019 No–15; 2022 No–15**

A company needs to acquire an equipment that will cost Tk. 3,35,200. The company has two alternatives to finance the requirement :

(a) To take lease from ILC for 5 years with an annual lease payment of Tk. 1,20,000 to be paid at the end of each period.

(b) To take loan from Jamuna Bank Ltd. @ 15% interest P.a. repayable annually at the end of each of the 5 years in equal installments. The equipment will be depreciated on straight line basis. Corporate tax rate is 40%. Which offer the company should accept?

Solution : ২২২ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

২০২৪ সালের পরীক্ষায় যা এসেছিল-

ⓍⓍ **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd 3rd (Old) 2009 ; (Marketing) 2021 No–15; 2024 No–16**

Mahi Co. Ltd./A company needs to acquire an equipment costing Tk. 5,00,000. The company has two alternatives to finance the equipment :

(i) To take lease from union capital for four years with an annual payment of Tk. 2,20,000 to be paid at the end of each year.

(ii) To take loan from AB bank @ 16% interest payable at the end of each year in equal installment for four years. The equipment will be depreciated on straight line basis. Corporate tax rate is 35%.

Which offer the company should accept?

Solution : ২২৪ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Finance 2018 No–15**

Mr. Saad is thinking of acquiring an equipment costing Tk. 15,00,000 and he has the choice of borrowing Tk. 15,00,000 at 15% to be repaid in 5 equal installment end of each year. It is assumed that the straight line method of depreciation is allowable under income straight line basis. The tax rate 40%. Which alternative is best for Mr. Saad?

Solution : ২২৮ পৃষ্ঠার **Problem-12** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 4th (Management) 2017 No–11.**

Shayla is thinking of acquiring a computer. She is in a position to decide whether the computer should be purchased by borrowing money from BRAC Bank at 12% or be acquired on lease rent basis. The price of the computer is Tk. 60,000. Its annual maintenance cost is Tk. 2,500 per year including in lease rental and expected life is 5 years. The computer will be depreciated on straight line basis. The residual value of the computer is Tk. 10,000. The rate of income tax is 40%.

In case of leasing, the lease rental is Tk. 20,000 per year for 5 years and maintenance expenses to be borne by lessor.

Which alternative is better for Shayla, leasing or buy borrowing?

Solution : ২৩২ পৃষ্ঠার **Problem-15** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 4th (Management) 2016 No–11.**

A Company wishes to acquire an asset costing Tk. 200 lakh. The company has an offer from a bank to lend @18% repayable in 3 equal year-end installments. A leasing company has also submitted a proposal to the company to acquire the asset on lease at yearly rentals of Tk. 450 per Tk. 1,000 of the asset value for 3 years payable at year-ends. The leasing company shall maintains the asset during lease period. The annual maintenance cost of the asset is estimated to Tk. 15 lakh. the rate of depreciation allowable for tax purposes is 50% on WDV. The expected salvage value of the asset is Tk. 35 lakh (net of tax). Tax rate is 40%. Which offer the company should accept.

Solution : ২৩৪ পৃষ্ঠার **Problem-16** নং প্রশ্নের উত্তর দেখুন।

Chapter 08 Operating and Financial Leverage

(পরিচালন ও অর্থায়ন লিভারেজ)

Part-A (Brief Questions) :

oo 01. **What is leverage?**

Ans : Leverage is the ability to use fixed cost fund with the aim of increasing shareholders income or, wealth.

oooo02. **When does leverage became zero?** 99%

[N.U. (Marketing)- 2016]

Ans : When fund from outside sources is not utilized, then the leverage becomes zero.

03. What is financial leverage? 99%

[N.U. (Marketing)- 2016, 2018, 2021, 2023]

Ans. Financial leverage is the ability to utilizes capital with fixed charges with the shareholders equity.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। –সম্পাদক]

04. What is meant by degree of operating leverage?

Ans : Degree of operating leverage means the percentage in EBIT to the percentage change in sales.

05. What is operating leverage? 99%

[N.U. (Marketing)- 2020]

Ans. Operating leverage is a measure of how a change in sales revenue leads to change in operating income, due to the presence of fixed operating costs.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। –সম্পাদক]

06. What do you mean by operating break-even point?

Ans : Operating break-even point is that sales level where total production cost equals to total sales

Part- B & C (Short & Broad Questions) :

01. What is financial leverage? [Ans মার্কেটিং ইন্সপেক্টর (২য় খণ্ড) বইয়ের 236 Page 8.02 No.]
02. Define leverage. **[Ans : 236 Page 8.01 No.]**
03. **"No fixed operating cost, no operating leverage"–comments.** 99% **[Ans : 238 Page 8.05 No.]**
04. What is meant by operating break-even-point? **[Ans : 238 Page 8.06 No.]**
05. Show the difference between operating break even point and financial break even point. **[Ans : 240 Page 8.10 No.]**
06. **Discuss the impact of financial leverage on cost of capital.** 99% **[Ans : 237 Page 8.03 No.]**

[N.U. (Marketing)- 2017]

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। –সম্পাদক]

[২০২৪ সালের পরীক্ষায় যা এসেছিল–

6. *Distinguish between operating leverage and financial leverage.* **[Ans : Page No. 239; Ques. No 8.07]**

[টেস্ট পরীক্ষা–লালমনিরহাট ও কুষ্টিয়া কলেজ; N.U. BBA (Hon's) (Marketing-3rd) 2016, 2018, 2024]

Or, Show the difference between operating leverage and financial leverage.

Or, Make difference between operating leverage and financial leverage.

Help Line ***Dear Students, In above Suggestions Part- B and C : 06 No. Questions in are most important for the Examinee of 2025.*** ***..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

Problem–01 ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2020 No–09; 2022 No–07**

Find the break even point (BEP) from the following information :

Selling price per unit	Tk. 100
Variable cost per unit	Tk. 50 and
Fixed operating cost is	Tk. 5,00,000

Solution : মার্কেটিং ইন্সপেক্টর (২য় খণ্ড) বইয়ের ২৪২ পৃষ্ঠার **Problem-01** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

০০ **Problem–02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2019 No–14**

The following data is extracted from AB Ltd.

Sales	30,000 units
Sales price	Tk. 25.00
Variable cost per unit	Tk. 12.50
Annual fixed cost	Tk. 3,00,000
Total assets	Tk. 10,00,000
10% debenture	Tk. 5,00,000
Share capital	Tk. 5,00,000
Corporate tax rate	25%

On the basis of the above data you are required to calculate :

(i) DOL,
(ii) DFL,
(iii) DTL,
(iv) BEP in units.

Solution : ২৪৩ পৃষ্ঠার **Problem-04** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

⊗⊗ **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2018 No-17; 2022 No-14; 2024 No–14**

Asha Co. Ltd./Vorosha Company Ltd. has supplied the following data :

Sales 10,000 units @ Tk. 40 per unit.
Variable cost per unit Tk. 25.
Fixed cost Tk. 80,000.
Interest on debenture Tk. 20,000.
You are required to calculate the following :
(i) DOL (ii) DFL (iii) DCL (iv) BEP (v) BEP Tk.

Solution : ২৪৪ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem–04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2021 No–12**

Jeny company manufactures and sells a single product. The company's sales and expenses for June 2022 are as follows :

Particulars	Total Tk.	Per unit Tk.
Sales	4,50,000	30.00
Less : Variable expenses	1,80,000	12.00
Contribution margin	2,70,000	18.00
Less : Fixed expenses	2,16,000	.....
Net income	54,000	.....

Requirements :

(i) What is the break-even point in unit sold?
(ii) What is the total contribution margin at break even point?
(iii) How many units would have to be sold each month to earn a target profit of Tk. 90,000?
(iv) Degree of operating leverage (DOL).

Solution : ২৪৫ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2017 No-14; 2023 No–11**

Following are the capital structure of Bata Shoe Company Ltd.
Common stock capital Tk. 2,00,000 @ Tk. 100 each
10% debenture of Tk. 1,00,000
Sales 30,000 units @ Tk. 10 per unit
Variable cost per unit Tk. 5 and fixed cost Tk. 80,000
Tax rate of the company is 30%
Calculate :
(i) DOL
(ii) DFL
(iii) DCL
(iv) What will be the effect of DOL, DFL and DCL on EBIT and EPS if sales increased by 40%? .

Solution : ২৪৬ পৃষ্ঠার **Problem-07** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

ooo **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Marketing 2018 No-16**

You are supplied with the following data relating to BRB Ltd. :

Variable expenses of sales	60%
Annual interest paid	Tk. 6,000
Degree of operating leverage	6 : 1
Degree of financial leverage	3 : 1
Income tax rate	30%

Compute earning after taxses of the company.

Solution : ২৪৮ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-07** ***Year Question & Solution*** ⊙ **N.U. (2nd) BBA (Hons.) Management 2021 No-14**

A Company's capital structure is composed of common stock capital of Tk. 2,00,000 (2,000 shares @ Tk. 100 each) and of 15% debenture of Tk. 3,00,000. Corporate tax rate of the company is 25%. The company's last year data as follows :
Sales 30,000 units @ Tk. 10 each
Variable cost @ Tk. 4 per unit
Fixed cost Tk. 80,000.
Requirement :
(i) What are the degrees of operating leverage, financial leverage and total leverage?
(ii) What will be the effect of DOL, DFL and DTL if sales increased by 40% and 60%?

Solution : ২৫১ পৃষ্ঠার **Problem-13** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 2nd Mgt 2019 No-16**

You are supplied with the following data relating to Tabassum Ltd. :

Sales	Tk. 8,00,000
Degree of operating leverage :	4 times
Degree of financial leverage :	2 times
EBIT	Tk. 50,000
Earning per share	Tk. 4

According to the above information, what will Tabassum's EBIT be if actually sales to be Tk.7,20,000? What will the EPS be? (ignoring taxation)

Solution : ২৫২ পৃষ্ঠার **Problem-14** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

⊗⊗ **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2024 No–09**

Calculate degree of operating leverage and degree of financial leverage from the following data :

Year	EBIT	Sales in units	EPS
2022	40,000	16,00,000	Tk. 20
2023	45,000	19,00,000	Tk. 25

Solution : ৫৩৯ পৃষ্ঠার **Question No-9** নং প্রশ্নের উত্তর দেখুন।

Chapter 09 Capital Structure Determination
(মূলধন কাঠামো নিরূপণ)

Part-A (Brief Questions) :

○○○01. **What is financial structure? (আর্থিক কাঠামো কি?)** 99%

Ans : The financial structure of a firm means the total of all liabilities and ownership claims the sum of what is usually the credit side of the balance sheet.

○ 02. **What is equation of financial structure? (আর্থিক কাঠামোর ইকুয়েশন কি?)**

Ans : Financial structure = Capital structure (Equity capital + Preferred stock + Long term debts) + Current liabilities.

○○○03. **What are the elements of capital structure? (মূলধন কাঠামোর উপাদান বলতে কি বুঝ?)**

Ans : The elements of capital structure are long term debt, preferred stock, common stock and retained earnings.

○○○04. **What is capital structure? (মূলধন কাঠামো কি?)**

Ans : Capital structure is the proportions of a firm's permanent long term finanang represented by debt, preferred stock, and common stock equity.

○○○05 **What is equity capital?** 99%

[N.U. (Marketing)- 2016, 2019, 2023]

Ans. Equity value is the value of a company's ownership interest, also known as equity capital.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○○06 **What is optimal capital structure? (কাম্য মূলধন কাঠামো কি?)** *[Test Exam : Eden, Uttara town, mohammadpur kenriya College; N.U. 2014, 2021]* 99%

Ans : Optimal capital structure is the capital structure that minimizes the firm's cost of capital and thereby maximizes the value of the firm.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○○07. **Define debt capital.** 99%

[D.U. (Affiliated) BBA(Hon's) (Marketing)- 2016]

Ans. Debt capital is mixture of debt and equity capital.

○○ 08. **What is trading on equity?**

(ব্যবসা ক্ষেত্রে নিজস্ব মূলধন কি?)

Ans : Trading on equity is the use of borrowed funds is expectation of higher return to equity holders.

○○○09 **What is Indifference points? (নিরপেক্ষ বিন্দু কি?)** 99%

[N.U. 2021]

Ans : Indifference points is the level of EBIT where EPS is the same of two alternatives.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○○10. **What do you mean by arbitrage process? (আর্বিট্রেজ প্রক্রিয়া বলতে কি বুঝ?)**

Ans : The arbitrage process is the purchase of securities whose prices are lower and sale of securities whose prices are higher, unrelated markets which are temporarily out of equilibrium.

○○○11. **What is Miller Model? (মিলার মডেল কি?)**

[Test Exam :Adoard College]

Ans : Merton Miller introduced a model designed to show how leverage affect firm's values when both personal and corporate taxes are taken into account.

○○○12 **What is stock dividend?** 99%

[N.U. (Marketing)- 2022]

Or, What do you mean by stock dividend?

Ans. When dividend is paid through bonus share it is called stock dividend.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে।"-সম্পাদক]

ooo13. What is debt capacity? (ঋণ ক্ষমতা কি?)

Ans : Debt capacity is the maximum proportion of debt that the firm can include in its capital structure and still maintain its lowest composite cost of capital.

Part- B & C (Short & Broad Questions):

oo 01. What is mean by capital structure? [Ans মার্কেটিং ইন্সপেক্টর (২য় খণ্ড) বইয়ের 255 Page 9.01 No.]

Or, Briefly explain the concept of capital structure. *[টেস্ট পরীক্ষা-ঢাকা ও এসকে কলেজ; N.U. BBS (Hon's) 2007]*

oo 02. Distinguish between capital structures and financial structure. **[Ans : 257 Page 9.09 No.]**

[টেস্ট পরীক্ষা- হাবিবুল্লাহ ও বরিশাল কলেজ]

ooo (03) **Differentiate between Net Income (NI) approach and net operating income (NOI) approach? 99%**

[Ans : 258 Page 9.10 No.] *[টেস্ট পরীক্ষা- ইডেন, কেসি, বরিশাল ও বরিশাল কলেজ;N.U. 2017 (Managment-4th) 2018; (Finance-3rd); D.U. 2016 (Finance-3rd)]*

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

ooo 04. What's meant by arbitrage? **[Ans : 256 Page 9.03 No.]**

ooo 05. **Explain the M.M. approach of the capital structure. 99% [Ans : 258 Page 9.12 No.]**

[টেস্ট পরীক্ষা- মুহসিন, তোলারাম ও বরিশাল কলেজ]

Help Line ***Dear Students, In above Suggestions Part- B and C : 03 & 05 No. Questions in are most important for the Examinee of 2025.*** ***..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

oo **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Finance 2016 No –4; 4th (Mgt.) 2018 No– 06**

The Ibnat company's/The ABC Ltd. expected annual net operating income (EBIT) is Tk. 50,000. The company has 10% debt Tk. 2,00,000. The equity capitalization rate of the company is 12.5%. The number of share of the company is 2,400.

Required :

(i) Find out the total value of the firm (V).

(ii) Find out the k_0 of the firm.

(iii) Determine the market price of per share.

Solution : মার্কেটিং ইন্সপেক্টর (২য় খণ্ড) বইয়ের ২৬৪ পৃষ্ঠার **Problem-03** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem–02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd Marketing 2021 No–08**

From the following data, determine the value of the firms belonging to same risk class, under NI approach :

Particulars	Levered firm-A	Unlevered firm-B
EBIT	Tk. 2,25,000	Tk. 2,25,000
Interest @ 15%	Tk. 75,000	Nil

Cost of equity capital is 20%.

Solution : ২৬৫ পৃষ্ঠার **Problem-04** নং প্রশ্নের উত্তর দেখুন।

○○○ Problem-03 *Year Question & Solution* ⊙ D.U. BBA (Hon's) 3rd Finance 2016 ; N.U. BBA (Hon's) 3rd (Old) 2012 No–8; 4th Year Management 2020 No–9

Reza Brother's Company/The ABIR and Co.s expected earning before interest and taxes (EBIT) is Tk. 50,000. The Co. has 10% debt of Tk. 2,00,000. The equity capitalization rate (k_e) of the company is 12.50%. The number of common stock of the company is 2,000.

Required :

(i) Find out the total value of the firm (V).
(ii) Find out the k_0 of the firm.
(iii) Determine the market value per share.
(iv) Determine the EPS.

Solution : ২৬৬ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

○○ Problem-04 *Year Question & Solution* ⊙ N.U. BBA (Hon's) 3rd (Old) 2014 No–12

Sun Limited with operating earning (EBIT) of Tk. 4,00,000 is attempting to evaluate a number of possible capital structures given below. Which of the capital structure will you recommend and why? According to net income approach :

Capital	Level of debt	Cost of debt	Cost of equity
1	2,00,000	11%	13%
2	3,00,000	11%	14.5%
3	4,00,000	12%	15.5%
4	5,00,000	13%	17%

Solution : ২৬৬ পৃষ্ঠার **Problem-07** নং প্রশ্নের উত্তর দেখুন।

○○ Problem-05 *Year Question & Solution* ⊙ N.U. BBA (Hon's) 3rd Finance 2018 No– 12

The following information is available for company 'K' and 'Q' : –

Details	K	Q
Net operating income	Tk. 2,00,000	Tk. 2,00,000
Interest on debt		Tk. 50,000
Cost of equity	16%	16%
Cost of debt	10%	10%

By using NI approach you are required the following questions :

(i) Market value of equity, market value of debt and value of both company.
(ii) What is the average cost of capital for each of the company?
(iii) What happens to the WACC of 'K' if it employees Tk. 3,00,000 of debt to finance a project that yields an operating income of Tk. 40,000?

Solution : ২৬৭ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

○○○ Problem-06 *Year Question & Solution* ⊙ N.U. BBA (Hon's) 3rd (Old) 2014 No–04

Green company presently is an unleveled firm. The company expects to generate Tk. 2,00,000 in earning before interest and taxes (EBIT), in perpetuity. The corporate tax rate is 35%. All earnings after taxes are paid as dividend. The firm is considering a capital structure to allow Tk. 1,00,000 of debt. It cost of debt is 12%. Unleveled firms in the same industry have a cost of equity capital of 18%. What will be the new value of Green company?

Solution : ২৬৮ পৃষ্ঠার **Problem-10** নং প্রশ্নের উত্তর দেখুন।

○○ Problem-07 *Year Question & Solution* ⊙ N.U. BBA (Hon's) 3rd (Old) 2015 No– 05

There are two companies U and L. Where L is levered firm and U is unleveled firm which has no debt. L firm has Tk. 1,50,000 debt at 10%. Assume that there is no corporate tax and personal tax. The expected EBIT is Tk. 5,00,000 and cost of equity is 20%

Required :

(a) Value of U firm (V_U)
(b) Value o L firm (V_L)
(c) Cost of equity of U firm
(d) Market value of share of L firm.

Solution : ২৬৯ পৃষ্ঠার **Problem-12** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Acc. 2016 No– 12(b)**

The following informations are given to you :

Details	Firm U	Firm L
EBIT	2,50,000	2,50,000
10% Debt		7,50,000
k_{eu}	15%	
Corporate tax rate	40%	40%
Personal tax rate	10%	10%

Required :

(a) Value of unlevered firm (V_u);
(b) Value of levered firm (V_L);
(c) Gain from leverage;
(d) Cost of equity of unlevercd firm;
(e) Cost of equity of levered firm (k_{el}).

Solution : ২৭০ পৃষ্ঠার **Problem-13** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old) 2009; 4th Year Management 2019 No–6**

ABC Company, the following debt and equity capital (after tax) have been made at various investment plans. You are required to determine optimum debt equity mix for the company by calculating the composite cost of capital :

Plan	% of debt	Cost of debt (k_d) (%)	Cost of equity (k_e) (%)
A	0	0	12
B	10	5	12
C	20	5.2	12.5
D	30	6	13
E	40	6.5	14
F	50	7	15
G	60	7.5	16

Solution : ২৭১ পৃষ্ঠার **Problem-14** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old) 2015 No–12; 4th Year Management 2018 No–12.**

PQR/ABC Ltd. has the following capital sturcture :-

Particulars	Amount
Equity (Tk. 100)	10,00,000
12% Debt	5,00,000
Total capital	15,00,000

The company wishes to raise Tk. 5,00,000 for expansion programms. The following alternatives are available :

(i) 100% equity.
(ii) 50% equity and 50% debt at 12%.
(iii) 100% debt at 12%.

The expected EBIT is Tk. 20,00,000 the tax rate is 40%. Calculate the EPS and which one would you prefer. Also calculate the indifference EBIT between alternatives *(i)* and *(ii)*.

Solution : ২৭৩ পৃষ্ঠার **Problem-16** নং প্রশ্নের উত্তর দেখুন।

Chapter 10 Introduction to Capital Market

(মূলধন বাজারের ভূমিকা)

Part-A (Brief Questions) :

○○○01. **What is financial market? (আর্থিক বাজার কি?)**

Ans : Financial markets are forums in which suppliers of funds and demanders of funds can transact business directly.

○○○02 **What is secondary market?(মাধ্যমিক বাজার কি?)** 99% *[N.U. 2021]*

Ans : Secondary market is the market in which existing, already outstanding securities are traded among investors. In secondary market, the original issuer has no part in the transaction.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

○○○03. **What is organized securities exchange? (সংগঠনের সিকিউরিটিজ এক্সচেঞ্জ কি?)**

Ans : The organized securities exchange is the formal organization that has physical location and exists to bring together buyers and sellers of securities in the secondary market.

○○○04. **What is best efforts arrangement? (সর্বোচ্চ প্রচেষ্টার ব্যবস্থা চুক্তি কি?)**

Ans : The best efforts arrangement is an arrangement for the sale of securities in which the investment bank handling the transaction gives no guarantee that the securities will be sold.

○○○05. **What is best efforts offering? (সর্বোচ্চ প্রচেষ্টা প্রস্তাব কি?)** 99% *[N.U. 2013]*

Ans : Best efforts offering instead of underwriting a securities issue, investment bankers may sell the issue on a best efforts offering basis.

○○ 06. **What is green shoe provision? (গ্রিন সু-প্রভিশন কি?)**

Ans : An option given to the underwriters to purchase additional number of shares is known as over allotment option or green shoe provision.

○○○07. **What is Initial Public Offering (IPO)? (প্রাথমিক পাবলিক প্রস্তাব/আইপিও কি?)**

[Test Exam : Edan, Azam Khan College]

Ans : When a company issues ordinary share to the public for the first time then it is known as initial public offering.

○○ 08. **What is long term financing? (দীর্ঘকালীন অর্থায়ন কি?)**

Ans : Long term financing is thought of as those which run for longer periods, typically for some period such as 10, 15, 20 years of more.

09. What is right share? (রাইট শেয়ার কাকে বলে?) [২০২৪ সালে যা এসেছিল] [N.U. 2016, 2018, 2022, 2024 (Marketing)]

***Ans :** The equity shares purchased by the existing shareholders under the pre-emptive right is termed as right share.*

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○○○11. **What is pre-emptive right?** 99%

(প্রি এমটিম আবিষ্কার কি?) *[N.U 2012]*

Ans : The privilege of the shareholders to maintain their proportional company ownership by purchasing a proportionate share of any new issue of common stock, or securities convertible into common stock.

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[N.U. (Marketing)-2016, 2020, 2024]

***Ans :** A market that enables suppliers and demanders of fund and demanders of long term funds to make transactions.*

○○○13. **What is blue sky law?** 99%

[D.U. (Affiliated) BBA(Hon's) (Marketing)- 2016]

Ans. Blue sky law is state regulations established as safeguards for investors against securities fraud.

○○○(14) **What is primary market? (প্রাথমিক বাজার কাকে বলে?)** 99% *[N.U. 2020]*

Ans : Primary market is the only market in which the corporate or government issuer is directly involved in transactions and receives direct benefit from the issue.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

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✪✪✪15. **What is public issue?** *[N.U. 2012]*
(পাবলিক ইস্যু কী?)
Ans: Public issue is the sale of bonds of stock to the general public.

✪ 16. **What is new issued market?**
(নতুন ইস্যুকৃত বাজার কি?)
Ans : The new issue market is the market for stock of companies that go public, and the issue is called an initial public offering (IPO).

✪✪✪17. **What is private placement?**
(ব্যক্তিগত উপস্থাপন কি?)
Ans : A private placement is the sale of stock to only one or a few investors, usually institutional investors.

✪✪✪18. **What is right offering? (রাইট প্রস্তাব কি?)**
Ans : A short-term option to buy a certain number (or fraction) of securities from the issuing corporation; also called a subscription right.

✪✪ 19. **Who are the insiders. (মালিক পক্ষ কারা?)**
Ans : The SEC also controls trading by corporate insiders, who are the officers, directors, and major stockholders of the firm.

20. What is zero coupon bond? [২০২৪ সালে যা এসেছিল] *[N.U. 2024]*
Or, Define zero coupon bond.
(জিরো কুপন বন্ডের সংজ্ঞা দাও।)
Ans : A zero-coupon is a bond bought at a price lower than its face value, with the face value repaid at the time of maturity.

✪✪✪21. **What is original discount bond?**
(প্রকৃত বাট্টা বণ্ড কি?)
Ans : An original issue discount (OID) bond is one which was issued at a price well below its par value. An OID bond can be either a zero coupon bond with an interest rate substantially lower than the current market rate.

✪✪✪22. **Define registration statement?**
(নিবন্ধন বিবরণী কি?)
Ans : The registration statement is a disclosure document filed with the SEC in order to register a new security and includes the prospectus and other SEC required information.

✪✪ 23. **What is the function of a mutual fund?**
(একটি মিউচ্যুয়াল ফান্ডের কাজ কি?)
Ans : A mutual fund sells shares to investors, pools the funds, and invests the funds in a portfolio of securities.

✪✪✪24. **What is BSEC? (BSEC কী?)** *[N.U. 2014]*
Ans: BSEC indicates Bangladesh Securities and Exchange Commission. It is the regulator of the capital market of Bangladesh Comprising DSE and CSE. The commission is a statutory body and attached to the ministry of finance.

✪✪✪25. **What do you mean by public offering?**
(পাবলিক প্রস্তাব বলতে কী বুঝ?) 99%
Ans : A public offering is the offering of securities of a company or a similar corporation to be public.

✪✪✪26. **What is right issue? (রাইট ইস্যু কী?)**
Ans : A right issue is an issue of rights to buy additional securities in a company made to the company's existing security holders.

✪✪✪27. **What is value of right? (রাইটের মূল্য কী?)**
Ans : Value of rights is the mathematically calculated value of a subscription right (a right to buy stock) after the offering is amounted but before the right expires.

✪✪✪28. **What is capital gain? (মূলধনী লাভ কী?)**
Ans : Capital gain is the amount by which the sale price of an asset exceeds the assets book price.

✪✪ 29. **What does convertible bond mean?**
(রূপান্তরযোগ্য বন্ড বলতে কী বুঝ?)
Ans : Convertible bond is that type of bonds which can be converted into common shares after a certain period of time.

30 **What is bond and debenture?**
(বন্ড ও ঋণপত্র কী?) 99%
[D.U. (Affiliated) BBA (Hon's) (Marketing)- 2016]
Ans. Bond : Bond is long term debt instrument issued by a government.
Debenture : Debenture refers to the unsecured bonds of a corporation.
রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক

✪✪✪31. **What is prospectus? (বিবরণপত্র কী?)**
Ans : A paper/ statement in which information about a new security issue and issuing company are state is called prospectus.

✪✪ 32. **What is preferred stock?**
(অগ্রাধিকার স্টক কী?)
Ans : Preferred stock is an equity security which may have any combination of features not possessed by common stock including features of both an equity and a debt instrument and is generally considered a hybrid instrument.

✪✪✪33. **What is bond refunding?** 99%
(বন্ড পুনঃতহবিলকরণ কী?)

Ans : Bonds that have their principal cash amount already held aside by the original issuer of the debt is called bond refunding.

✪✪✪34. **What is convertible preferred stock?**
(রূপান্তরযোগ্য অগ্রাধিকার শেয়ার কী?)

Ans : These are preferred issues which holders can exchange for predetermined number of the company's common stock shares.

✪✪✪35. **What is money market? (মুদ্রা বাজার কি?)**

Ans : A money market is a financial market for issuing and trading short-term debt securities.

Part- B & C (Short & Broad Questions) :

✪✪ 01. What is mean by pre-emptive right?
[Ans : বেসিক মার্কেটিং ইজ্‌প্রেস (২য় খণ্ড) বইয়ের 283 Page 10.27 No.]

✪✪✪ 02. Discuss the IPO process. **[Ans : 279 Page 10.09 No.]** *[N.U. 2013, 2015; D.U. 2016 (Accouting-3rd)]*
Or, What are the initial public offering?

✪✪✪ 03. **Show the differences between bond and common stock.** 99% **[Ans : 281 Page 10.19 No.]**
[টেস্ট পরীক্ষা–আযম খান ও এমএম কলেজ; N.U. 2012, 2014; 2016 (Marketing-3rd); 2017 (Accouting-3rd); D.U. 2017 (Accouting-3rd)]
Or, Distinguish between common stock and bond.

✪✪✪ 04. **Show the differences between right issue and public offering.** 99% **[Ans : 277 Page 10.02 No.]**
[টেস্ট পরীক্ষা– এমএম, বিএল, রাজশাহী, দিনাজপুর ও এমএম কলেজ; N.U 2013, 2015; 2017 (Accouting-3rd); D.U. 2017 (Accouting-3rd)

✪✪ 05. Discuss the classification in bond in brief. **[Ans : 284 Page 10.28 No.]** *[N.U. 2014, 2015]*

✪✪✪ 06. What is meant by share market? Explain the classification of share market. *[N.U. (Marketing)- 2017]*

✪✪ 07. What is zero coupon bonds? Why do investors buy it although it has no coupon interest. *[N.U. 2015]*

✪✪✪ 08. What is bond refunding? Why is it done? **[Ans : 278 Page 10.05 No.]** *[N.U. 2014, 2015]*

✪✪✪ 09. **Briefly explain underwriting as a method of public offering.** 99% **[Ans : 278 Page 10.06 No.]**
[N.U. 2012, 2014; 2018 (Managment-3rd)]

✪✪ 10. Why debt is the cheapest source of fund? *[N.U. 2019 (Marketing-3rd)*

✪✪✪ (11) **What is capital market? Discuss the types of capital market.** 99%
[Ans : 277+282 Page 10.01+10.21 No.] *[N.U. 2019 (Marketing-3rd)]*
[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

✪✪✪ (12) **Show the differences between money market and capital market.** 99% **[Ans : Page No 281; Ques. No 10.17]**
[টেস্ট পরীক্ষা–উত্তরা টাউন ও হরগঙ্গা কলেজ;N.U. 2018 (Accouting-3rd); 2016 (Finance-3rd); 2017, 2022 (Marketing-3rd)]
Or, Distinguish between the money and capital market.
[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

✪✪✪ 13. **(a) Discuss the methods of selling share of a company to the publics.** 99%
[Ans : Page No 284; Ques. No 10.30] *[N.U. 2020, 2022 (Marketing-3rd)]*
Or, What are the methods of selling share of a company to the publics?
(b) Explain the rights of the ordinary stockholder. 99% **[Ans : Page No 278; Ques. No 10.03]**
[N.U. 2020, 2022 (Marketing-3rd)]
[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

✪✪✪ (14) **Describe the role of Bangladesh Securities and Exchange Commission (BSEC) as a controller of capital market.** 99% **[Ans : Page No 283; Ques. No 10.25]** *[N.U. 2016, 2019, 2023 (Marketing-3rd)]*
[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

Help Line ***Dear Students, In above Suggestions Part- B and C : 03, 11, 12 & 14 No. Questions in are most important for the Examinee of 2025.*** ***..... Editor***

গাণিতিক সমস্যাবলি ও সমাধান (Mathematical Problems & Solution)

Easy Math Basic Problem : Category-B & C

○○ **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 4th Year (Mgt) 2020 No–6**

X Ltd. is considering a rights offer. The company has determined that ex-rights price would be Tk. 74. Subscription price is Tk. 44. One new share of stock can be purchased for each five shares currently owned. You are required to calculate market price share.

Solution : মার্কেটিং এক্সপ্রেস (২য় খণ্ড) বইয়ের ২৮৬ পৃষ্ঠার **Problem-03** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 4th Year (Mgt) 2019 No–8**

Mr. John has 10,000 equity shares of a company. Recently the company declared a right offering which 5,000 new shares to be issued at Tk. 10 each. At present the company has 1,00,000 shares in market at Tk. 50 per share. How many shares will be able to purchase by Mr. John under propose right offer?

Solution : ২৮৭ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem–03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hons.) 3rd (Old) 2013 No–06; Marketing 2021 No–09**

The stock of the Ocean company is selling for Tk. 150 per share. The company issues rights that allow the holders to subscribe for one additional share of stock at a subscription price of Tk. 125 a share for each nine rights held. Compute the theoretical value of the following :

(a) A right when the stock is selling "rights-on;"

(b) One share of stock when it goes "ex-rights."

Solution : ২৮৭ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Old) 2012 No–13; 4th Year Management 2018 No–16**

Mitu Cotton Ltd./Moni Cotton Ltd. has 50,000 shares outstanding and each share is currently selling at Tk. 40. For the proposed R and D program the company needs to acquire and fund of Tk. 5,00,000. The financial manager of the company, Mr. Hossain wants to raise the required fund by offering right shares at Tk. 25 per share.

Required :

(i) No. of new issued share.

(ii) No. of rights needed to buy a new share.

(iii) Value of right before ex-right.

Solution : ২৯০ পৃষ্ঠার **Problem-10** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc.) 2016 No–14; 2020 No–05**

Hilary Inc. is proposing a rights offering. Presently they have 2,40,000 shares outstanding at Tk. 80 each. There will be 60,000 new shares offered at Tk. 60 each.

Required :

(a) What is the new market value of the company?

(b) How many rights needed to purchase one new share?

(c) What is the ex-right price?

(d) What is the value of a right?

(e) Why might a company have a right offerings rather than a general cash offer?

Solution : ২৯২ পৃষ্ঠার **Problem-13** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Finance 2016 No–14**

Saniat corporation is planning to market a new product. To finance the venture is proposed to make a right issue at Tk. 10 of one new share for each two shares held. The company currently has outstanding 1,00,000 shares priced at Tk. 40 per share. Assuming that the new money is invested to earn a fair return, find the values for the following :

(i) Number of new issued shares.
(ii) Number of rights needed to buy one share.
(iii) Amount of new investment.
(iv) Total value of the company after issue.
(v) Total number of shares after issue.

Solution : ২৯৪ পৃষ্ঠার **Problem-16** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Finance) 2020 No–16**

ACI Ltd. currently has 10,00,000 shares outstanding at current market price of Tk. 120 per share. The company needs Tk. 2,00,00,000 for proposed project. The board of director of the company has decided to collect fund by issuing right share. The subscription price has been set at Tk. 70 per share.

Calculate :

(a) Number of new shares to be required to collect fund
(b) Number of right
(c) The price of share after right issue
(d) The price of ex-right
(e) The price of share under right on.

Solution : ২৯৭ পৃষ্ঠার **Problem-19** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd Marketing 2016 No–17; 4th Year Management 2020 No–14**

You have 10 shares of Ahmed corporation/Nafiu Wasiu Corporation. The current market price of the stock is Tk. 50. You also have Tk. 80 in cash. You have just received a news of rights offering from the company. One new share can be purchased at Tk. 40 for each five shares currently owned by you.

Calculate :

(a) What is the value of a right?
(b) What is the value of share at ex-right?
(c) What is the value of your assets before right offering?
(d) What will be the value of your assets if you sell the rights?
(e) What will be the value of your assets if you do not exercise your rights?

Solution : ২৯৯ পৃষ্ঠার **Problem-22** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Acc) 2019 No–14**

The Big Boss Corporation has announced a rights offer to raise Tk. 30,00,000 for a new journal, the journal of financial express. This journal will review potential articles after the author pays a non refundable reviewing fee of Tk. 3,000 per page. The stock is selling Tk. 60 per share currently and there are 2,40,000 shares outstanding.

(i) If the subscription price is set Tk. 50 per share, how many shares must be sold?
(ii) How many rights will it take to buy one share?
(iii) What is the ex-right price?
(iv) What is the value of a right?
(v) Show how a shareholder with 1,000 shares before the offering and no desire to buy additional shares is not harmed by the right offer.

Solution : ৩০১ পৃষ্ঠার **Problem-23** নং প্রশ্নের উত্তর দেখুন।

গ্রন্থ সহায়ক : উপরিউক্ত সাজেশন্সের পূর্ণাঙ্গ উত্তরপত্র পেতে.....
BP BATICKOM PUBLICATION ব্যাতিক্রম মার্কেটিং ইন্সপেক্স (২য় খণ্ড) বইটি সংগ্রহ করুন।

জাতীয় বিশ্ববিদ্যালয়ের প্রশ্নাবলি ও উত্তরপত্র

Financial Management (In English); Subject Code : 232305

National University

[BBA (Hon's) 3rd Year; Exam-2019 (Held on-09/02.2020)]

(Department of Marketing)

Subject Code : 232305

Subject Name : Financial Management (In English)

Time : 4 hours; **Full marks :** 80

[**N.B.** Different parts of each question should be answered chronologically]

Group-A

1. Answer any **ten** questions. $1 \times 10 = 10$

(a) What is wealth maximization?

Ans. Wealth maximization means maximizing the net present value or wealth of a course of action for the shareholders.

(b) What is agency problem?

Ans. The probability of conflict of interest between the shareholders and management of a firm is known as agency problem.

(c) What is annuity due?

Ans. Annuity due is an annuity where the cash flow occurs at the beginning of the every period.

(d) What is the profitability index (PI)?

Ans. Profitability index be defined as the ratio which is obtained by dividing the sum of present value of future cash inflows by the present value of the cash outlay.

(e) What is ratio analysis?

Ans. Ratio analysis is the process of determining and interpreting numerical relationships based on financial statements. Ratio analysis is used to evaluate various aspects of a company's operating and financial performance such as its efficiency, liquidity, profitability and solvency.

(f) What is rule of 72?

Ans. Rule of 72 gives a good approximation of the interest rate or, the number of years required to double the principal amount normally in case of annually compounding. $i = \frac{72}{n}$ and $n = \frac{72}{i}$

(g) What is zero coupon bond?

Ans. The bond which is bright at a price lower than its face value, with the face value repaid at the time of maturity is called zero coupon bond.

(h) What is WACC?

Ans. Weighted average cost of capital (WACC) is the minimum rate of return a firm must earn on its assets to satisfy its investors.

(i) What is factor?

Ans. Any number that is finitely divisible by that number is called is factor.

(j) What is equity capital?

Ans. Equity value is the value of a company's ownership interest, also known as equity capital.

(k) What is capital rationing?

Ans. The Capital Rationing refers to the choice of investment proposals under financial constraints in terms of a given size of capital.

(l) What is uncertainty?

Ans. Uncertainty refers to the outcomes of given event which are too unsure to be assigned probabilities.

Group-B

[Answer any **five** questions.] $4 \times 5 = 20$

2. Discuss the goal and objectives of a financial manager.
[Ans : বিস্তারিত মার্কেটিং ইঞ্জিনিয়ার্স (২য় খণ্ড) বইয়ের Page No. 7 Ques. No. 1.10]
3. Write down the reasons of time preference for money. ***[Ans : Page No 18; Ques. No 2.08]***
4. Why debt is the cheapest source of fund?
5. Discuss the types of capital market. ***[Ans : Page No 282; Ques. No 10.21]***
6. How much you have to deposit now to get Tk. 6,00,000 after 20 years at 8 percent interest per annum?
Solution : ***Page No-21; Problem No-02.***
7. Troll Ltd. has total sales of the year Tk. 6,90,000, cost of goods sold Tk. 6,55,000, Accounts receivable Tk. 70,000, Inventory Tk. 2,00,000.
Calculate the following ratios :
(i) Inventory turnover ratio
(ii) Accounts receivable turnover
(iii) Average collection period
Solution : ***Page No-100; Problem No-02.***
8. If a company has 30% debt and 70% equity and earns 15% on its assets before taxes. What is the percent of return on equity, assuming it pays 10% for debt capital while tax rate is 50%?
9. What is agency problem? Discuss in brief the various techniques of overcoming agency problem.
[Ans : Page No 9+10; Ques. No 1.16+1.17]

Group-C

[Answer any **five** questions.] $5 \times 10 = 50$

10. (a) Discuss the principles of business finance. ***[Ans : Page No. 5 Ques. No. 1.04]***
(b) Discuss the three major decisions that financial manager must take. ***[Ans : Page No 8; Ques. No 1.13]***
11. You have a plan to accumulate Tk. 12,00,000 ten years from now for buying a car. You are deciding to deposit a certain amount of money in a special account that will provide you 9.5% interest per annum. What will be your annual deposit at the beginning of each year?
Solution : ***Page No-38; Problem No-38.***
12. You are considering purchase of an apartment at Dhanmondi costing Tk. 1,00,00,000. You have three offers :
(i) Pay full in cash
(ii) Pay 25% in cash immediately and pay Tk. 13,20,000 each year for the next 10 years.
(iii) Pay 50% in cash and take a loan of Tk. 50,00,000 from BHBFC at 15% interest to be repaid in equal monthly instalment over the next 20 years. Assume 12% opportunity cost.
Which offer should you accept?
Solution : ***Page No-44; Problem No-43.***
13. From the following data of machine-A; you are required to calculate IRR :

		Tk.
	Machine cost	60,000
Annual estimated income after depreciation and income tax		
Year:	1	6,300
	2	5,400
	3	7,300
	4	10,000
	5	12,000
		41,000

Estimated life 5 years
Estimated salvage value 3,000
Average income tax rate 40%
Depreciation has been charged on straight line basis. Cost of capital is 12%.
Solution : ***Page No-76; Problem No-23.***

14. The following data is extracted from AB Ltd.

Sales	30,000 units
Sales price	Tk. 25.00
Variable cost per unit	Tk. 12.50
Annual fixed cost	Tk. 3,00,000
Total assets	Tk. 10,00,000
10% debenture	Tk. 5,00,000
Share capital	Tk. 5,00;000
Corporate tax rate	25%

On the basis of the above data you are required to calculate : (i) DOL (ii) DFL (iii) DTL (iv) BEP in units.

Solution : ***Page No-243; Problem No-04.***

15. A company needs to acquire an equipment that will cost Tk. 3,35,200. The company has two alternatives to finance the requirement :
   (a) To take lease from United Leasing Company for 5 years with an annual lease payment of Tk. 1,20,000 to be paid at the end of each period.
   (b) To take loan from IFIC bank at 15% interest p. a. repayable annually at the end of the 5 years in equal installments. The equipment will be depreciated on straight line basis. Corporate tax rate is 40%.

   Which offer the company should accept?

Solution : ***Page No-222; Problem No-08.***

16. Describe the role of Bangladesh Securities and Exchange Commission (BSEC) as a controller of capital market. ***[Ans : Page No 283; Ques. No 10.25]***

17. Afroza Company needs Tk. 1,60,000 to meet working capital requirement. It has three alternative sources :
   (i) The company can buy Tk. 2,00,000 of materials on terms of 3/30, net 90.
   (ii) The company can take advance of Tk. 2,00,000 at 13% interest rate with 20% compensating balance requirement.
   (iii) A factor will buy company's accounts receivable and will advance at 12% interest p. a. upto 80% of the face value. Annual credit sales Tk. 12,00,000 and average collection period is 60 days. The factor will charge 2% commission. It has been estimated that there will be administration cost of Tk. 1,000 per month and 1% bad debt loss.

   Required : Which alternative should the company choose?

Solution : ***Page No-199; Problem No-51.***

National University

[BBA (Hon's) 3rd year; Exam-2020 (Held on-12/03/2022)]

(Department of Marketing)

Subject Code : 232305

বিষয় : Financial Management (In English)

Time : 4 hours **Full marks :** 80

[N.B. Answer different questions of each group chronologically.]

Group-A

1. Answer any **ten** questions. $1 \times 10 = 10$

(a) What is financial management?

Ans. Financial management means planning, organizing, directing and controlling the financial activities such as procurement and utilization of funds of the enterprise.

(b) What is portfolio investment?

Ans. Portfolio investment refers to investments made in a group of assets, including transactions in equity securities, such as common stock, and debt securities, such as banknotes, bonds and debentures.

(c) What is perpetuity?

Ans. A perpetuity is a security that providesa constant stream of identical cash flows indefinitely with no end.

(d) What is rule of 69?

Ans. Rule of 69 gives a good approximation of the interest rate or, the number of years required to double the principal amount normally in case of annual compounding. $I = 0.35 + \frac{69}{n}$ $n = 0.35 + \frac{69}{i}$

(e) What is payback period?

Ans. The payback period is the time required for the firm to recover its initial investment in a project, as calculated from cash flows.

(f) What is current ratio?

Ans. The current ratio is a liquidity ratio that measures whether a firm has enough resources to meet its short-term obligations.

(g) What is primary market?

Ans. primary market is the only market in which the corporate or government issuer is directly involved in transactions and receives direct benefit from the issue.

(h) What is trade credit?

Ans. The credit which is created by purchasing of goods and services on credit is called trade credit.

(i) What is lease financing?

Ans. Lease financing is the process by which a firm can obtain the use of certain fixed assets for which it must make a series of contractual, periodic, tax deductible payments.

(j) What is operating leverage?

Ans. Operating leverage is a measure of how a change in sales revenue leads to change in operating income, due to the presence of fixed operating costs.

(k) What is working capital?

Ans. Working capital simply refers to the capital which is required to finance current assets.

(l) What is capital market?

Ans. A market that enables suppliers and demanders of fund and demanders of long term funds to make transactions.

Group-B

[Answer any **five** questions.] $4 \times 5 = 20$

2. Does a financial manager have any social responsibility? If so, discuss.
[Ans : বিস্তারিত মার্কেটিং ইনসাইট (২য় খণ্ড) বইয়ের Page No. 9 Ques. No. 1.14]

3. What rate is required for a sum of money to be triple after 5 years?
Solution : ***Page No-25; Problem No-09.***

4. Which is the best method of evaluation of project out of net present value (NPV) and internal rate of return (IRR)?

5. The following are the information of Aurin Foods limited—
Inventory turnover 6 times, Average collection period 40 days, Average payment period 30 days, Assume 1 year = 360 days.
Calculate : (i) Operating cycle; (ii) Cash conversion cycle.
Solution : ***Page No-154; Problem No-02.***

6. Write down the assumption of explicit and implicit cost. ***[Ans : Page No 119; Ques. No 5.13]***

7. CPC Food Ltd. cash requirement Tk. 40,00,000 next year. The cost of fund is 10% per annum. The company has transaction cost of Tk. 100 per transaction.
Requirement :
(i) Optimum cash balance;
(ii) The total annual cost of demand for optimum cash balance.
Solution : ***Page No-159; Problem No-11.***

8. Show the difference between lease financing and hire purchase. ***[Ans : Page No 211; Ques. No 7.2.11]***

9. Find the break even point (BEP) from the following information :

Selling price per unit	Tk. 100
Variable cost per unit	Tk. 50 and
Fixed operating cost is	Tk. 5,00,000

Solution : ***Page No-242; Problem No-01.***

Group-C

[Answer any **five** questions.] $5 \times 10 = 50$

10. "Wealth maximization, net the profit maximization is the ultimate objective of firm."—Explain.
[Ans : Page No. 11 Ques. No. 1.21]

11. Explain the significance of financial management in business. *[Ans : Page No 7; Ques. No 1.09]*

12. NEXUS company is considering a project which requires an investment of Tk. 5,00,000. The estimates salvage value is Tk. 50,000 after 5 years of estimated life. Tax rate is 40%. The company uses straight line method of depreciation and the project has the following proposed cash flows before tax (CFBT) :

Year.	CFBT (Tk.)
1	2,50,000
2	1,50,000
3	1,50,000
4	1,00,000
5	1,00,000

Determine : (1) Payback period, (2) Average rate of return and (3) Net present value @ 12% rate.
Solution : *Page No-79; Problem No-26.*

13. Aurni Fashion House gives you the following information :

Particulars	Taka
Merchandise inventory	1,20,000
Note receivable	30,000
Cash in hand	45,000
Notes payable	20,000
Accounts receivable	60,000
Cash at bank	90,000
Accounts payable	1,00,000
Accrued expenses	10,000
Unearned revenue	5,000

Requirements: (i) Gross working capital (ii) Net working capital (iii) Current ratio (iv) Acid test ratio.
Solution : *Page No-102; Problem No-05.*

14. The capital structure of Wasi Company is as follows :

	Taka
8% debenture	1,00,000
12% preference share	50,000
Common shares	1,20,000
Retained earnings	30,000
Total	3,00,000

The common shares of the company sell for Tk. 30 each. It is expected that the company will pay next year a dividend of Tk. 2.5 per share that will grow at 4% forever. The market price of preference share is Tk. 96 (per value Tk. 100). A flotation cost of 5% of the market price would be incurred to the issue of all kinds of shares. The company's marginal tax rate is 40%. Calculate the cost of capital of Wasi Company.
Solution : *Page No-129; Problem No-15.*

15. Assume that a firm has owner's equity of Tk. 1,00,000. The ratios for the firm are :

Current debt to total debt	0.40
Total debt to owner's equity	0.60
Fixed assets to owner's equity	0.60
Total assets turnover	2 times
Inventory turnover	8 times

Complete the following balance sheet, given the information above.

Liabilities and equities	Tk.	Asset	Tk.
Current debt	–	Cash	–
Long terms debt	–	Inventory	–
Total debt	–	Total current assets	–
Owner's equity	–	Fixed assets	–
Total capital	–	Total assets	–

Solution : *Page No-109; Problem No-12.*

16. Raihan Company Ltd. has the following capital structure :

Particulars	Taka
Common stock (50,000 shares)	50,00,000
Retained earnings	30,00,000
12.5% preferred stock	20,00,000
14.5% debentures	20,00,000
Total capital	1,20,00,000

Additional information :

(i) The current market price of the share is Tk. 150 per share. Current dividend per share is Tk. 20 per share which is expected to grow at 9% per year and flotation cost of common stock is 4.5% of current market price.

(ii) The face value of preferred stock is Tk. 100 per share of which current market price is Tk. 120 and flotation cost is Tk. 8 per share. Maturity period is 5 years.

(iii) The par value of debenture is Tk. 100, which is sold for Tk. 90 and flotation cost is 5% of face value. Maturity period is 10 years.

(iv) The corporate tax rate is 40%.

Required : Find out the WACC using book value weight.

Solution : *Page No-128; Problem No-14.*

17. (a) What are the methods of selling share of a company to the publics? *[Ans : Page No 284; Ques. No 10.30]*

(b) Explain the rights of the ordinary stockholder.

National University

[BBA (Hon's) 3rd year; Exam-2021 (Held on-11/05/2023)]

(Department of Marketing)

Subject Code : 232305

Subject : Financial Management (In English)

Time : 4 hours **Full marks :** 80

[N. B. Answer different questions of each group chronologically.]

Group-A

1. Answer any ten questions. $1 \times 10 = 10$

(a) What is wealth maximization?

Ans. Wealth maximization means maximizing the net present value or wealth of a course of action for the shareholders.

(b) What is business risk?

Ans. Business risk refers to the risk associated with the firm's operations its Impact is shown in the varibility of firms operating income (EBIT).

(c) What is IRR?

Ans. The Internal Rate of Return is the discount rate that equates the sum of the present values of series cash inflows with the initial investment.

(d) What is effective annual interest rate?

Ans. An effective annual interest rate is the real return on a savings account or any interest paying investment when the effect of compounding over time are taken into accounts.

(e) What is optimum capital structure?

Ans. Optimal capital structure is the capital structure that minimizes the firm's cost of capital and thereby maximizes the value of the firm.

(f) **What is financial leverage?**

Ans. Financial leverage is the ability to utilizes capital with fixed charges with the shareholders equity.

(g) **What is EPS?**

Ans. Earning per share is a financial metric that measures the portion of a company's profit that is allocated to each outstanding share of common stock.

(h) **What is capital budgeting?**

Ans. Capital budgeting is the process of identifying, analyzing and selecting investment projects whose returns (cash flows) are expected to extend beyond one year.

(i) **What is secondary market?**

Ans. Secondary market is the market in which existing, already outstanding securities are traded among investors. In secondary market, the original issuer has no part in the transaction.

(j) **What is floating cost?**

Ans. The cost that is occurred in issuing securities is called floating cost.

(k) **What is annuity?**

Ans. When equal amount of money is paid or, received continuously with equal time interval is called annuity.

(l) **What is indifference points?**

Ans. Indifference points is the level of EBIT where EPS is the same of two alternatives.

Group-B

[Answer any **five** questions.] $4 \times 5 = 20$

2. Discuss the goal and objectives of a financial manager.

[Ans মার্কেটিং ইন্টারেক্টস (২য় খণ্ড) বইয়ের **Page No. 7 Ques. No. 1.10]**

3. Mr. Alam deposited Tk. 5,00,000 in a bank. The bank promised to give 12% interest per annum. What will be the terminal value after 7 years if compounded annually?

Solution : *Page No-21; Problem No-01.*

4. Show the differences between operating lease and financial lease. *[Ans : Page No 211; Ques. No 7.2.10]*

5. Cost of equipment Tk. 5,00,000. Expected life 5 years, salvage value at the end of life Tk. 50,000, annual CFBT Tk. 1,80,000, corporate tax rate 40% calculate PBP. Comment on the result.

Solution : *Page No-62; Problem No-07.*

6. From the following information calculate (i) Current ratio and (ii) Acid test ratio :

Assets	Tk.	Liabilities	Tk.
Cash	20,000	Accounts payable	20,000
Accounts receivables	50,000	Expense payable	30,000
Inventory	70,000	Long-term loan	1,25,000
Prepaid expense	20,000	Capital	60,000
Fixed assets	75,000		
Total	2,35,000		2,35,000

Solution : *Page No-101; Problem No-04.*

7. Discuss the importance of capital budgeting. *[Ans : Page No 52; Ques. No 3.03]*

8. From the following data, determine the value of the firms belonging to same risk class, under NI approach :

Particulars	Levered firm-A	Unlevered firm-B
EBIT	Tk. 2,25,000	Tk. 2,25,000
Interest @ 15%	Tk. 75,000	Nil

Cost of equity capital is 20%.

Solution : *Page No-265; Problem No-04.*

9. The stock of the Ocean company is selling for Tk. 150 per share. The company issues rights that allow the holders to subscribe for one additional share of stock at a subscription price of Tk. 125 a share for each nine rights held. Compute the theoretical value of the following :

(a) A right when the stock is selling "rights-on."

(b) One share of stock when it goes "ex-rights."

Solution : *Page No-287; Problem No-06.*

Group-C

[Answer any **five** questions.] $10 \times 5 = 50$

10. (a) Why wealth maximization is considered as the ultimate goal of the firm? Explain.

[Ans : Page No. 11 Ques. No. 1.21]

(b) Discuss the major decisions taken by a financial manager. ***[Ans : Page No 8; Ques. No 1.13]***

11. You are considering of an apartment in Dhaka costing Tk. 95,00,000.
You have three offers :
(i) Pay full in cash.
(ii) Pay 30% in cash immediately and pay Tk. 9,30,000 at the end of the each year for 10 years.
(iii) Pay 50% in cash and take a loan of Tk. 45,00,000 from AB Bank Ltd. @ 15% p.a. for 20 years. Assume the opportunity cost 12%. Which offer should you accept?

Solution : ***Page No-45; Problem No-44.***

12. Jeny company manufactures and sells a single product. The company's sales and expenses for June 2022 are as follows :

Particulars	Total Tk.	Per unit Tk.
Sales	4,50,000	30.00
Less : Variable expenses	1,80,000	12.00
Contribution margin	2,70,000	18.00
Less : Fixed expenses	2,16,000	.....
Net income	54,000	.....

Requirements :
(i) What is the break-even point in unit sold?
(ii) What is the total contribution margin at break even point?
(iii) How many units would have to be sold each month to earn a target profit of Tk. 90,000?
(iv) Degree of operating leverage (DOL). **Solution :** ***Page No-245; Problem No-06.***

13. The capital structure of XYZ company is as follows :

Particulars	Tk.
8% Debenture	1,00,000
12% Preference shares	50,000
Common shares	1,20,000
Retained earnings	30,000
Total	3,00,000

The common shares of the company sells for Tk. 30 each. It is expected that the company will pay next year a dividend of Tk. 2.5 per share that will grow at 4% forever. The market price of preference share is Tk. 95 (per value Tk. 100). A floatation cost of 5% of the market price would be incurred to the issue of all kinds of shares. The company's marginal tax rate is 40%. Compute the cost of capital of XYZ company.

Solution : ***Page No-131; Problem No-16.***

14. Zaria Ltd. needs Tk. 5,00,000 to finance its intermediate term capital requirement for remodeling of product processing equipments. Sonali Bank Ltd. has agreed to provide loan at 14% interest. The loan has to be paid in next 5 years. You are required to show the payment schedule under :
(a) Balloon payment method
(b) Capital recovery method.

Solution : ***Page No-207; Problem No-07.***

15. Mahi Co. Ltd. needs to acquire a new machine costing Tk. 5,00,000.
There are two alternative way of financing :
(i) To take loan from NCC Bank @ 16% interest payable at the end of each year in equal installment for four years.
(ii) To take lease from union capital for four years with an annual payment of Tk. 2,20,000 to be paid at the end of each year. Which offer the company should accept? Assume the corporate tax rate is 35% and the SLM of depreciation is applicable.

Solution : ***Page No-224; Problem No-09.***

16. ABC company is considering a project which requires an investment of Tk. 4,00,000. The estimated salvage value is zero. Tax rate is 50% and cost of capital is 12%. The company uses straight line depreciation and the proposed project cash flows before tax (CFBT) are as follows :

Year	CFBT
1	Tk. 1,00,000
2	Tk. 1,00,000
3	Tk. 1,50,000
4	Tk. 1,50,000
5	Tk. 2,50,000

Required : (a) NPV (b) IRR. **Solution :** ***Page No-78; Problem No-25.***

17. Sara Ltd. requires working capital of Tk. 20 lakh. There are three alternatives :
    (i) Trade credit : Terms of sale 2/20, net 90.
    (ii) Bank loan : Trust bank is interested to grant Tk. 20,00,000 as a loan @ 10% per annum after reduction of 25% as compensating balance.
    (iii) Commercial paper : To issue a commercial paper @ 10%. The issuing cost would be Tk. 20,000 in each quarter.
    Which option the company should accept? **Solution :** ***Page No-197; Problem No-49.***

National University

[BBA (Hon's) 3rd Year; Exam-2022 (Held in-02/05/2024]

(Department of Marketing)

Subject Code : 232305

Subject : Financial Management (In English)

Time : 4 hours; **Full marks :** 80

[**N.B** Answer different questions of each group chronologically.]

Group-A

1. Answer any **ten** questions. Marks : 1 × 10 = 10

(a) What is financial management?

Ans. Financial management means planning, organizing, directing and controlling the financial activities such as procurement and utilization of funds of the enterprise.

(b) What is corporate social responsibility?

Ans. Corporate social responsibility is a business outlook that acknowledges a firms responsibilities to its stakeholders and natural environment.

(c) What is compound interest?

Ans. Compound interest is the interest calculated on the principal and the interest accumulated over the previous period.

(d) What is right share?

Ans. The equity shares purchased by the existing shareholders under the pre-emptive right is termed as right share.

(e) What is stock dividend?

Ans. When dividend is paid through bonus share it is called stock dividend.

(f) What is working capital?

Ans. Working capital simply refers to the capital which is required to finance current assets.

(g) What is trade credit?

Ans. The credit which is created by purchasing of goods and services on credit is called trade credit.

(h) What is pay back period?

Ans. The payback period is the time required for the firm to recover its initial investment in a project, as calculated from cash flows.

(i) What is cost of capital?

Ans. Cost of capital is minimum required rate of return.

(j) What is CAPM?

Ans. The capital asset pricing model is a model that describes the relationship between the expected return and risk of investing in a security.

(k) What is annuity due?

Ans. An annuity whose payments occurs at the beginnings of each period is called an annuity due.

(l) What is lease financing?

Ans. Lease financing is the process by which a firm can obtain the use of certain fixed assets for which it must make a series of contractual, periodic, tax deductible payments.

Group-B

[Answer any **five** question.] Marks : 4 × 5 = 20

2. Differentiate between wealth maximization and profit maximization.
[Ans মার্কেন্টিং ইন্টারেক্ট (২য় খণ্ড) বইয়ের **Page No. 10; Ques. No 1.19]**

3. Calculate effective interest rate (EIR) for Tk. 5,000 invested in a project at 10% interest monthly.
Solution : ***Page No-30; Problem No-20.***

4. XYZ Company paid Tk. 12 as dividend per share of common stock last year. Each share of Tk. 100 is currently selling at a premium of 5%. Dividend would grow by 5% per annum and flotation cost is 5%. Determine the cost of common stock. **Solution :** ***Page No-120; Problem No-01.***

5. In what time a sum of money will be doubled at 5% interest according to the rule of 69?
Solution : ***Page No-26; Problem No-12.***

6. Jabber Ltd. has total sales of the year Tk. 6,90,000, Cost of goods sold Tk. 6,55,000, Accounts receivable. Tk. 70,000, Inventory Tk. 2,00,000.
Calculate the following :
(i) Inventory turnover ratio
(ii) Accounts receivable turnover ratios
(iii) Average collection period. **Solution :** ***Page No-100; Problem No-02.***

7. Find the break even point (BEP) from the following information :
Selling price per unit Tk. 100
Variable cost per unit Tk. 50
Fixed operating cost is Tk. 5,00,000. **Solution :** ***Page No-242; Problem No-01.***

8. ABC Company extended credit facility for Tk. 10,00,000 on term of (i) 2/15 net 45 (ii) 2/20 net 90. Calculate which alternative is appropriate for the company. **Solution :** ***Page No-185; Problem No-32.***

9. Show the differences between money market and capital market. ***[Ans : Page No 281; Ques. No 10.17]***

Group– C

[Answer any **five** question.] Marks : 10 × 5 = 50

10. You have a plan to accumulate Tk. 12,00,000 ten years from now for buying a car. You are deciding to deposit a certain amount of money in special account that will provide you 9% interest per annum.
(i) What will be your annual deposit at the end of each year?
(ii) What will be your annual deposit at the beginning of each year? **Solution :** ***Page No-37; Problem No-37.***

11. Dema Ltd. is considering a proposal to install a new machine. The project will cost Tk. 1,00,000 and its expected life is 5 years. The estimated cash inflows before tax (EFBT) of the project as follows :

Year	(Tk.) EFBT
1	25,000
2	30,000
3	35,000
4	40,000
5	45,000

Requirements : (a) PBP; (b) NPV at 10%; (c) IRR. **Solution :** ***Page No-81; Problem No-28.***

12. ABC Company was recently formed to manufacture a new product. It has the following capital structure :

Account titles	Taka
14% Debenture	12,00,000
12% Preferred stock	4,00,000
Common stock	16,00,000

The common stock sells for Tk. 50 and is expected to pay a Tk. 4 dividend this year, which will grow at 10%. Corporate tax rate is 50%. Compute weighted average cost of capital. **Solution :** ***Page No-125; Problem No-11.***

13. The following balance sheet have been taken from the books of Muhsinat and Co. :

Liabilities	Taka	Assets	Taka
Share capital	1,00,000	Fixed costs	1,19,000
Retained earnings	30,000	Inventories	1,20,000
Debenture	1,25,000	Debtors/Accounts receivable	90,000
Creditors	75,000	Cashable securities	8,000
Expense payable	15,000	Prepaid expense	10,000
Income tax payable	5,000	Cash	3,000
	3,50,000		3,50,000

You are required to compute : (i) Working capital; (ii) Current ratio; (iii) Quick ratio; (iv) Proprietary ratio.
Solution : ***Page No-104; Problem No-07.***

14. Asha Co. Ltd has supplied the following data :
Sales 10,000 units @ Tk. 40 per unit.
Variable cost per unit Tk. 25.
Fixed cost Tk. 80,000
Interest on debenture Tk. 20,000.
You are required to calculate the following : (i) DOL; (ii) DFL ; (iii) DCL ; (iv) BEP (Tk.)
Solution : ***Page No-244; Problem No-05.***

15. A company needs to acquire an equipment that will cost Tk. 3,35,220. The company has two alternatives to finance the equipment :
(a) To take lease from Popular Leasing Company for 5 years with an annual lease payment of Tk. 1,20,000 to be paid at the end of each period.
(b) To take loan from Modhumoti Bank Ltd. at 15% interest p.a. repayable annually at the end of the 5 years in equal installments. The equipment will be depreciated on straight line basis. Corporate tax rate is 40%.
Which offer the company should accept? **Solution :** ***Page No-222; Problem No-08.***
Note : সমাধানকৃত অংকের সাথে ২০২২ সালের অংকটির পার্থক্য হলো Equipment cost Tk. 3,35,200 টাকার জায়গায় 3,35,220 টাকা দেওয়া আছে। ২০২২ সালের অংকটি সমাধানের সময় Equipment cost Tk. 3,35,220 ধরে সমাধান করলেই হবে।

16. (a) Discuss the methods of selling share of a company to the publics. ***[Ans : Page No 284; Ques. No 10.30]***
(b) Explain the rights of the ordinary stockholder.

17. (a) Discuss the importance of financial management. ***[Ans : Page No 7; Ques. No 1.09]***
(b) Describe the scope of financial management. ***[Ans : Page No 7; Ques. No 1.08]***

National University

[BBA (Hon's) 3rd Year; Exam.-2023 (Held in-30/04/2025)]

(Department of Marketing)

Subject Code : 232305

Subject : Financial Management (In English)

Time : 4 hours; **Full marks :** 80

[N.B. Answer different questions of each group chronologically.]

Group-A

1. Answer any **ten** questions. Marks : 1 × 10 = 10

(a) What is wealth maximization?

Ans. Wealth maximization means maximizing the net present value or wealth of a course of action for the shareholders.

(b) What is ratio analysis?

Ans. Ratio analysis is the process of determining and interpreting numerical relationships based on financial statements. Ratio analysis is used to evaluate various aspects of a company's operating and financial performance such as its efficiency, liquidity, profitability and solvency.

(c) What is WACC?

Ans. Weighted average cost of capital (WACC) is the minimum rate of return a firm must earn on its assets to satisfy its investors.

(d) What is equity capital?

Ans. Equity value is the value of a company's ownership interest, also known as equity capital.

(e) What is IRR?

Ans. The internal rate of return is the discount rate that equates the sum of the present values of series cash inflows with the initial investment.

(f) What is financial leverage?

Ans. Financial leverage is the ability to utilizes capital with fixed charges with the shareholders equity.

(g) What is floating cost?

Ans. The cost that is occurred in issuing securities is called floating cost.

(h) What is rule of 72?

Ans. Rule of 72 gives a good approximation of the interest rate or, the number of years required to double the principal amount normally in case of annually compounding. $i = \frac{72}{n}$ and $n = \frac{72}{i}$

(i) What is capital budgeting?

Ans. Capital budgeting is the process of identifying, analyzing and selecting investment projects whose returns (cash flows) are expected to extend beyond one year.

(j) What is annuity?

Ans. When equal amount of money is paid or, received continuously with equal time interval is called annuity.

(k) What is business risk?

Ans. Business risk refers to the risk associated with the firm's operations its Impact is shown in the varibility of firms operating income (EBIT).

(l) What is operating lease?

Ans. Operating lease is a cancelable contractual arrangement whereby the lessee agrees to make periodic payments to the lessor.

Group-B

[Answer any **five** questions.] Marks : 4 × 5 = 20

2. Discuss the goal and objectives of a financial manager.

[Ans : মার্কেটিং ইনসাইট (২য় খণ্ড) বইয়ের Page No. 7; Ques. No 1.10]

3. In what time a sum of money will be doubled at 10% interest according to the rule 69?
Solution : *Page No-27; Problem No-13.*

4. You have a plan to buy a motorcycle 5 years from now. The price of the motorcycle will be Tk. 5,00,000. You want to accumulate this money in a bank account by monthly installment, which gives you 10% interest. What is your monthly installment?
Solution : *Page No-34; Problem No-29.*

5. Show the difference between lease and purchase. *[Ans : Page No 213; Ques. No 7.2.16]*

6. Mr. Ashraf has taken a loan of Tk. 10,00,000 from a bank. After 10 years he has given the bank Tk. 17,00,000. Calculate the rate of interest if interest compounded quarterly.
Solution : *Page No-26; Problem No-11.*

7. Current market price is Tk. 300 per share, current dividend is Tk. 12 per share. Dividend per share is expected to increase at about 2% per share. Calculate the cost of equity share capital.
Solution : *Page No-121; Problem No-02.*

8. Discuss the importance of capital budgeting. *[Ans : Page No 52; Ques. No 3.03]*

9. Cost of equipment Tk. 5,00,000. Expected life 5 years, salvage value at the end of life Tk. 50,000, annual CFBT Tk. 1,80,000, corporate tax rate 40%. Calculate PBP and comment on the result.
Solution : *Page No-62; Problem No-07.*

Group-C

[Answer any **five** questions.] Marks : 10 × 5 = 50

10. (a) Discuss the principles of business finance. *[Ans : Page No. 5; Ques. No 1.04]*
(b) Discuss the three major decisions that financial manager must take. *[Ans : Page No 8; Ques. No 1.13]*

11. Following are the capital structure of Bata Shoe Company Ltd.
Common stock capital Tk. 2,00,000 @ Tk. 100 each
10% debenture of Tk. 1,00,000
Sales 30,000 units @ Tk. 10 per unit
Variable cost per unit Tk. 5 and fixed cost Tk. 80,000
Tax rate of the company is 30%
Calculate : (i) DOL (ii) DFL (iii) DCL.
Solution : *Page No-246; Problem No-07.*

12. Mubira Ltd. needs Tk. 5,00,000 to finance its intermediate term capital requirement for remodeling of product processing equipments. Rupali Bank Ltd. has agreed to provide loan at 14% interest. The loan has to be paid in next 5 years. You are required to show the payment schedule under :
(a) Balloon payment method;
(b) Capital recovery method.
Solution : *Page No-207; Problem No-07.*

13. A chemical company is considering a project which requires an investment of Tk. 4,00,000. The estimates salvage value is zero. Tax rate is 50%. The company uses straight line depreciation and the proposed project has cash flows before tax (CFBT) as follows :

Year	CFBT
1	1,00,000
2	1,00,000
3	1,50,000
4	1,50,000
5	2,50,000

Determine :
(i) Payback period
(ii) Average rate of return
(iii) Net present value @ 15% PV factor.
Solution : *Page No-77; Problem No-24.*

14. ABC company gives you the following information :

Accounts titles	Tk.
Merchandise inventory	1,20,000
Notes receivable	30,000
Cash in hand	45,000
Notes Payable	20,000
Account receivable	60,000
Cash at bank	90,000
Accounts payable	1,00,000
Accrued expenses	10,000
Unearned revenue	5,000

You are required to calculate :

(i) Gross working capital,
(ii) Net working capital,
(iii) Current ratio,
(iv) Acid test ratio.

Solution : ***Page No-102; Problem No-05.***

15. Your brother committed to present you a personal computer on your graduation after 5 years from now. The computer will cost Tk. 50,000 at that time. Your brother is planning to accumulate the fund by depositing equal amount of money in a bank account bearing 8% interest rate per annum. How much should your brother deposit in the bank account if the deposits are made (i) yearly, (ii) quarterly?

Solution : ***Page No-31; Problem No-23.***

16. Abul Khair Company is thinking to purchase a machine for Tk. 12,290. The company can borrow Tk. 12,290 at an interest rate of 10% to be repaid in 10 equal annual installment of Tk. 2,000 or it can buy the machine for an annual lease payment of Tk. 2,000. Under income tax act straight-line lease of depreciation is allowed. The company pays tax @ 50%. Which alternative should the company accept?

Solution : ***Page No-219; Problem No-06.***

17. Describe the role of Bangladesh Securities and Exchange Commission (BSEC) as a controller of capital market.

[Ans : Page No 283; Ques. No 10.25]

National University

[BBA (Hon's) 3rd Year; Exam.-2024 (Held in-04/01/2026)]

(Department of Marketing)

Subject Code : 232305

Subject : Financial Management (In English)

Time : 4 hours; **Full marks :** 80

[N.B. Answer different questions of each group chronologically.**]**

Group -A

1. Answer any **ten** questions : Marks $1 \times 10 = 10$

(a) What is financial management?

Ans : Financial management means planning, organizing, directing and controlling the financial activities such as procurement and utilization of funds of the enterprise.

(b) What is rule of '69'?

Ans : Rule of 69 gives a good approximation of the interest rate or, the number of years required to double the principal amount normally in case of annual compounding. $1 = 0.35 + \frac{69}{n}$ $n = 0.35 + \frac{69}{i}$

(c) What is trade credit?

Ans : The credit which is created by purchasing of goods and services on credit is called trade credit.

(d) What is right share?

Ans : The equity shares purchased by the existing shareholders under the pre-emptive right is termed as right share.

(e) What is CAPM?

Ans. The capital asset pricing model is a model that describes the relationship between the expected return and risk of investing in a security.

(f) What is capital rationing?

Ans : The capital rationing refers to the choice of investment proposals under financial constraints in terms of a given size of capital.

(g) What is agency problem?

Ans : The probability of conflict of interest between the shareholders and management of a firm is known as agency problem.

(h) What is zero coupon bond?

Ans : A zero-coupon is a bond bought at a price lower than its face value, with the face value repaid at the time of maturity.

(i) What is portfolio investment?

Ans. Portfolio investment refers to investments made in a group of assets, including transactions in equity securities, such as common stock, and debt securities, such as banknotes, bonds and debentures.

(j) What is current ratio?

Ans. The current ratio is a liquidity ratio that measures whether a firm has enough resources to meet its short-term obligations.

(k) What is working capital?

Ans : Working capital simply refers to the capital which is required to finance current assets.

(l) What is capital market?

Ans : A market that enables suppliers and demanders of long term funds to make transactions, is called capital market.

Group -B

[Answer any **five** questions.] Marks : $4 \times 5 = 20$

2. Make differences between wealth maximization and profit maximization.

[Ans : মার্কেটিং হাইলাইটস (২য় খণ্ড) বইয়ের Page No. 10; Ques. No 1.19]

3. Zabeer Company extended credit facility for Tk. 10,00,000 on terms of (i) 2/15, net 45 (ii) 2/20, net 90. Calculate which alternative is appropriate for the company. **Solution :** *Page No-185; Problem No-32.*

4. Jamuna Company paid Tk. 12 as dividend per share of common stock last year. Each share of Tk. 100 is currently selling at a premium of 5%. Dividend would grow by 5% per annum and flotation cost is 5%. Determine the cost of common stock. **Solution :** *Page No-120; Problem No-01.*

5. A Company has the following information :

Accounts titles	Taka
Accounts receivable	40,000
Cash	10,000
Bank overdraft	10,000
Inventory	1,00,000
Accounts payable	40,000

Calculate the following ratios and comment on the solvency position and working capital :

(i) Current ratio

(ii) Acid test ratio

(iii) Working capital ratio. **Solution :** *Page No-106; Problem No-09.*

6. Distinguish between operating leverage and financial leverage. ***[Ans : Page No. 239; Ques. No 8.07]***

7. Lotus Company has a revolving credit of Tk. 5,00,000 at an interest of 13% with Agrani Bank PLC. Average borrowing of the company was Tk. 3,00,000 for the year. A commitment fee of 2.5% is to be paid. Compute EIR? **Solution :** ***Page No-538; Question No-07***

8. Cash inflow after tax (CFAT) of Project-A of BD company are :

Year	0	1	2	3	4	5
Project-A	(11,111)	4,444	3,333	9,999	5,555	9,789

Calculate payback period (PBP).

Solution : ***Page No-539; Question No-08***

9. Calculate degree of operating leverage and degree of financial leverage from the following data :

Year	EBIT	Sales in units	EPS
2022	40,000	16,00,000	Tk. 20
2023	45,000	19,00,000	Tk. 25

Solution : ***Page No-539; Question No-09***

Group -C

[Answer any **five** questions.] Marks : 10 × 5 = 50

10. (a) Discuss brief the internal factors affecting financial decisions. ***[Ans : Page No. 13; Ques. No 1.30]***

    (b) Discuss the scope of financial management. ***[Ans : Page No. 7; Ques. No 1.08]***

11. Padma Ltd is considering a proposal to install a new machine. The project will cost Tk. 1,00,000 and its expected life is 5 years. The estimated cash inflows before tax (CFBT) of the project as follows :

Year	CFBT (Tk.)
1	25,000
2	30,000
3	35,000
4	40,000
5	45,000

Requirements :

(a) PBP; (b) ARR; (c) NPV at 10%; (d) IRR; (e) Surplus life. **Solution :** ***Page No-81; Problem No-28.***

12. Jawwad Company has the following current assets and liabilities :

Accounts titles	Taka
Inventory	1,70,000
Cash	22,000
Accounts receivable	45,000
Bank overdraft	20,000
Accounts payable	45,000
Prepaid expense	12,000
Accrued salaries	22,000

During the year sales were Tk. 8,00,000 and purchase were Tk. 4,40,000. Determine the following ratios :

(a) Current ratio

(b) Acid test ratio

(c) Inventory turnover ratio

(d) Accounts receivables and payables turnover ratio. **Solution :** ***Page No-103; Problem No-06.***

13. Discuss the factors to be considered in determining working capital. ***[Ans : Page No. 143; Ques. No 6.11]***

14. Vorosha Company Ltd. has supplied the following data :

Sales 10,000 units @ Tk. 40 per unit.

Variable cost per unit Tk. 25.

Fixed cost Tk. 80,000.

Interest on debenture Tk. 20,000.

You are required to calculate the following :

(i) DOL (ii) DFL (iii) DCL (iv) BEP (v) BEP Tk. **Solution :** ***Page No-244; Problem No-05.***

15. You are considering purchase of an apartment at Gulshan costing Tk. 1,00,00,000. You have three offers :

(i) Pay full in cash.

(ii) Pay 25% in cash immediately and pay Tk. 10,00,000 each year for the next 10 years.

(iii) Pay 50% in cash and take a loan of Tk. 50,00,000 from Sonali Bank PLC at 12% interest to be repaid in equal monthly installment over the next 15 years. Assume 10% opportunity cost.

Which offer should you accept?

Solution : ***Page No-541; Question No-15***

16. A company needs to acquire an equipment costing Tk. 5,00,000. The company has two alternatives to finance the equipment :

(i) To take lease from union capital for four years with an annual payment of Tk. 2,20,000 to be paid at the end of each year.

(ii) To take loan from AB bank @ 16% interest payable at the end of each year in equal installment for four years. The equipment will be depreciated on straight line basis. Corporate tax rate is 35%.

Which offer the company should accept? **Solution :** ***Page No-224; Problem No-09.***

17. The capital structure of Bangladesh Food Ltd. are as follows :

Accounts titles	Taka
Common stock (Tk. 100 per share)	20,00,000
14% Preferred stock (Tk. 1,000 per share)	10,00,000
15% Debentures (Tk. 2,500 per debenture)	15,00,000
Retained earnings	5,00,000
	50,00,000

Other information :

(i) Current market price of common stock in Tk. 150. Current year company declares a dividend of Tk. 15 per share. The company expects that this dividend will increase at 10% annual rate forever.

(ii) The company preferred stock is redeemable after 5 years at 10% premium. The flotation cost of preferred stock is 20% of sales price. Currently the preferred stock sales at Tk. 1,200.

(iii) The maturity period of debentures is 10 years. After 10 years debentures is redeemable at 12% premium. But currently sales at 5% discount. The flotation cost of debenture is 1.5% of its face value. The corporate tax rate 40% and personal tax rate 25%.

You are required to calculate the overall cost of capital. **Solution :** ***Page No-127; Problem No-13.***